CHAPTER I

Observables and Pure States

1 The Structure of Algebras of Observables

1.1 Jordan–Lie Algebras and C^*-Algebras

In this section we specify the key algebraic and functional-analytic structures relevant to classical and quantum mechanics. Our main aim is to look at a C^*-algebra from the point of view of its self-adjoint part. From this perspective the relationship between the respective algebraic structures of classical and quantum mechanics is particularly transparent.

Recall that an **algebra** is a vector space with a (not necessarily commutative or associative) bilinear and distributive multiplication $\circ$. We write $A^2 := A \circ A$.

Definition 1.1.1. *A (real)* **Jordan algebra** *is a (real) algebra where*

$$A \circ B = B \circ A; \tag{1.1}$$

$$A \circ (B \circ A^2) = (A \circ B) \circ A^2. \tag{1.2}$$

The simplest motivation for (1.2) is that it is automatically satisfied when the Jordan product $\circ$ comes from an associative product via $A \circ B = \frac{1}{2}(AB + BA)$. However, not all Jordan algebras arise in this way.

Theories of dynamical significance have a second algebraic operation.

Definition 1.1.2. *A* **Jordan–Lie algebra** *is a real vector space $\mathfrak{A}_{\mathbb{R}}$ equipped with two bilinear maps $\circ$ and $\{\ ,\ \}$ (referred to as the* **Jordan product** *and the* **Poisson bracket***, respectively), such that the following conditions are satisfied. Firstly, one has*

$$A \circ B = B \circ A;$$

$$\{A, B\} = -\{B, A\} \tag{1.3}$$

for all $A, B \in \mathfrak{A}_{\mathbb{R}}$. *Secondly, for each* A, *the map* $B \mapsto \{A, B\}$ *is a derivation of* $(\mathfrak{A}_{\mathbb{R}}, \circ)$ *as well as of* $(\mathfrak{A}_{\mathbb{R}}, \{\ ,\ \})$. *This means that the* **Leibniz rule**

$$\{A, B \circ C\} = \{A, B\} \circ C + B \circ \{A, C\} \tag{1.4}$$

as well as the **Jacobi identity**

$$\{A, \{B, C\}\} = \{\{A, B\}, C\} + \{B, \{A, C\}\} \tag{1.5}$$

must hold for all $A, B, C \in \mathfrak{A}_{\mathbb{R}}$. *Finally, for all* $A, B, C \in \mathfrak{A}_{\mathbb{R}}$ *and some* $\hbar^2 \in \mathbb{R}$ *one requires the* **associator identity**

$$(A \circ B) \circ C - A \circ (B \circ C) = \tfrac{1}{4}\hbar^2\{\{A, C\}, B\}. \tag{1.6}$$

A Jordan–Lie algebra in which $\circ$ *is associative is called a* **Poisson algebra**.

It follows from these axioms that $(\mathfrak{A}_{\mathbb{R}}, \circ)$ is a real Jordan algebra, whereas $(\mathfrak{A}_{\mathbb{R}}, \{\ ,\ \})$ is a real Lie algebra. In connection with Jordan–Lie algebras, the terminology (non) associative always refers to the Jordan product.

The following definitions are recorded for later use.

Definition 1.1.3. *A* **Jordan morphism** *between Jordan–Lie algebras* $\mathfrak{A}_{\mathbb{R}}$ *and* $\mathfrak{B}_{\mathbb{R}}$ *is a linear map* $\beta : \mathfrak{A}_{\mathbb{R}} \to \mathfrak{B}_{\mathbb{R}}$ *satisfying* $\beta(A \circ B) = \beta(A) \circ \beta(B)$ *for all* $A, B \in \mathfrak{A}_{\mathbb{R}}$. *Similarly, a* **Poisson morphism** *between such algebras is a linear map satisfying* $\beta(\{A, B\}) = \{\beta(A), \beta(B)\}$. *A map between Jordan–Lie algebras that is simultaneously a Jordan morphism and a Poisson morphism is called simply a* **morphism**. *An invertible (Jordan, Poisson) morphism* $\alpha : \mathfrak{A}_{\mathbb{R}} \to \mathfrak{A}_{\mathbb{R}}$ *is called a* **(Jordan, Poisson) automorphism**, *and an invertible (Jordan, Poisson) morphism* $\alpha : \mathfrak{A}_{\mathbb{R}} \to \mathfrak{B}_{\mathbb{R}}$ *is a (Jordan, Poisson)* **isomorphism**.

We now equip the algebras introduced above with a norm.

Definition 1.1.4. *A* JB**-algebra** *is simultaneously a real Jordan algebra and a Banach space in which for all* $A, B \in \mathfrak{A}_{\mathbb{R}}$ *one has*

$$\|A \circ B\| \leq \|A\|\,\|B\|; \tag{1.7}$$

$$\|A\|^2 \leq \|A^2 + B^2\|. \tag{1.8}$$

The motivation for the axioms of a JB-algebra will emerge in due course. Putting $A = B$ in (1.7) and $B = 0$ in (1.8), one sees that given (1.7), axiom (1.8) is equivalent to the pair

$$\|A^2\| = \|A\|^2; \tag{1.9}$$

$$\|A^2\| \leq \|A^2 + B^2\|. \tag{1.10}$$

Definition 1.1.5. *A* JLB**-algebra** *is a* JB*-algebra* $\mathfrak{A}_{\mathbb{R}}$ *equipped with a Poisson bracket that makes it a Jordan–Lie algebra for some* $\hbar^2 \geq 0$.

A JLB-algebra with $\hbar = 0$ may alternatively be regarded as a Poisson algebra with zero Poisson structure. A Poisson algebra with nonzero Poisson bracket cannot, in general, be normed in such a way that the bracket is defined on the norm-completion of the algebra. For this reason Poisson algebras are usually not studied in the setting of Banach spaces.

A JLB-algebra $\mathfrak{A}_{\mathbb{R}}$ turns out to be the real part of a complex associative algebra $\mathfrak{A}$ of a much-studied type.

An **involution** on a complex algebra is a real-linear map $A \mapsto A^*$ such that for all $A, B \in \mathfrak{A}$ and $\lambda \in \mathbb{C}$ one has

$$A^{**} = A; \tag{1.11}$$
$$(AB)^* = B^*A^*; \tag{1.12}$$
$$(\lambda A)^* = \overline{\lambda}A^*. \tag{1.13}$$

A ***-algebra** is a complex associative algebra with an involution.

Definition 1.1.6. *A* C^***-algebra** *is a complex Banach space* $\mathfrak{A}$ *that is at the same time a* **-algebra, such that for all* $A, B \in \mathfrak{A}$ *one has*

$$\|AB\| \leq \|A\| \; \|B\|; \tag{1.14}$$
$$\|A^*A\| = \|A\|^2. \tag{1.15}$$

Combining the two axioms for a C^*-algebra leads to $\|A\| \leq \|A^*\|$; replacing A by A^* and using (1.11) yields

$$\|A^*\| = \|A\|. \tag{1.16}$$

It can actually be shown that (1.15) implies (1.14), but this highly nontrivial fact distracts from the guiding idea that a C^*-algebra is a specialization of a **Banach algebra**. This is a complex Banach space and an associative algebra, in which all A, B satisfy (1.14). This property guarantees that left and right multiplication are bounded, hence continuous; in fact, multiplication is a jointly continuous operation. For example, the space $\mathfrak{B}(\mathcal{B})$ of all linear maps on a Banach space $\mathcal{B}$ is a Banach algebra under the norm

$$\|A\| := \sup\{\|A\Psi\| \mid \Psi \in \mathcal{B}, \; \|\Psi\| = 1\}. \tag{1.17}$$

The C^*-axioms are motivated by the following example. Consistent with the above terminology, a ***-algebra of bounded operators** on some Hilbert space $\mathcal{H}$ is a collection of bounded operators on $\mathcal{H}$ that is closed under addition, scalar multiplication, operator multiplication, and taking adjoints. Thus the role of the involution is played by the adjoint. Recall the definition of the norm of a bounded operator:

$$\|A\|^2 := \sup\{(A\Psi, A\Psi) | \; \Psi \in \mathcal{H}, \; (\Psi, \Psi) = 1\}. \tag{1.18}$$

Since in a Hilbert space the norm is defined by $\|\Psi\|^2 = (\Psi, \Psi)$, eq. (1.18) is evidently a special case of (1.17). Hence $\|A\Psi\| \leq \|A\| \; \|\Psi\|$, which implies (1.14). Moreover, we estimate

$$\|A\Psi\|^2 = (A\Psi, A\Psi) = (\Psi, A^*A\Psi) \leq \|A^*A\| \; \|\Psi\|^2,$$

so that $\|A\|^2 \leq \|A^*A\|$, which is $\leq \|A^*\| \, \|A\|$ by (1.14). This leads to (1.16) by the argument preceding that equation; the ensuing inequality $\|A^*A\| \leq \|A\|^2$ then implies (1.15).

Definition 1.1.7. *A* **morphism** *between C^*-algebras $\mathfrak{A}$, $\mathfrak{B}$ is a linear map $\varphi : \mathfrak{A} \to \mathfrak{B}$ such that*

$$\varphi(AB) = \varphi(A)\varphi(B); \tag{1.19}$$

$$\varphi(A^*) = \varphi(A)^* \tag{1.20}$$

for all $A, B \in \mathfrak{A}$. An **isomorphism** *is a bijective morphism. Two C^*-algebras are* **isomorphic** *when there exists an isomorphism between them.*

It is clear that a C^*-algebra morphism between $\mathfrak{A}$ and $\mathfrak{B}$ restricts to a morphism (in the sense of 1.1.3) between the associated JLB-algebras $\mathfrak{A}_{\mathbb{R}}$ and $\mathfrak{B}_{\mathbb{R}}$ (cf. 1.1.9), and vice versa. Morphisms between C^*-algebras have excellent properties; see 1.3.10. For example, an isomorphism is automatically isometric.

Theorem 1.1.8. *A norm-closed *-algebra $\mathfrak{A}$ in $\mathfrak{B}(\mathcal{H})$ is a C^*-algebra (with operator multiplication as the product, etc.). Conversely, any C^*-algebra is isomorphic to a norm-closed *-algebra in $\mathfrak{B}(\mathcal{H})$, for some Hilbert space $\mathcal{H}$.*

The computation following (1.18) establishes the first half. The proof of the converse will be given at the end of 1.5. □

An element A of a *-algebra $\mathfrak{A}$ for which $A^* = A$ is called **self-adjoint**. The **self-adjoint part** $\mathfrak{A}_{\mathbb{R}}$ is the collection of all self-adjoint elements in $\mathfrak{A}$, seen as a real vector space. Since one may write

$$A = A' + iA'' := \tfrac{1}{2}(A + A^*) + i\tfrac{-i}{2}(A - A^*), \tag{1.21}$$

every element of $\mathfrak{A}$ is a linear combination of two self-adjoint elements.

A **commutative C^*-algebra** is a C^*-algebra in which the associative multiplication is commutative. The connection between JB-algebras, Jordan–Lie algebras, and C^*-algebras is as follows.

Theorem 1.1.9. *Let $\mathfrak{A}$ be a C^*-algebra, and choose $\hbar \in \mathbb{R}\backslash\{0\}$. Equipped with the norm inherited from $\mathfrak{A}$, and the operations*

$$A \circ B := \tfrac{1}{2}(AB + BA);$$

$$\{A, B\}_\hbar := \frac{i}{\hbar}[A, B], \tag{1.22}$$

the self-adjoint part $\mathfrak{A}_{\mathbb{R}}$ of $\mathfrak{A}$ is a JLB-algebra. When $\mathfrak{A}$ is noncommutative, the parameter $\hbar$ in (1.6) equals $\hbar$ in (1.22); in particular, one has $\hbar^2 > 0$. When $\mathfrak{A}$ is commutative, $\mathfrak{A}_{\mathbb{R}}$ is a Poisson algebra with zero Poisson bracket.

Conversely, given a JLB-algebra $\mathfrak{A}_{\mathbb{R}}$ for which $\hbar^2 \geq 0$, its complexification $\mathfrak{A}$ is a C^*-algebra under the operations

$$AB := A \circ B - \tfrac{1}{2}i\hbar\{A, B\}; \tag{1.23}$$

$$(A + iB)^* := A - iB; \tag{1.24}$$

$$\|A\| := \|A^*A\|^{\frac{1}{2}}. \tag{1.25}$$

In (1.24) we assume that $A, B \in \mathfrak{A}_{\mathbb{R}}$, and concerning (1.25) we remark that $A^*A \in \mathfrak{A}_{\mathbb{R}}$ for any $A \in \mathfrak{A}$.

To prove the first half of the theorem, first note that by (1.16) the involution in $\mathfrak{A}$ is continuous, so that $\mathfrak{A}_{\mathbb{R}}$ is a closed subspace of $\mathfrak{A}$. The axioms for a JLB-algebra are trivially verified, except (1.10). We defer the proof of this property to the end of 1.4.

In the opposite direction, it is trivially verified that the product (1.23) is associative as a consequence of the properties of a Jordan–Lie algebra. When (1.25) defines a norm, the property (1.15) holds by construction.

When $\mathfrak{A}_{\mathbb{R}}$ is associative, so that $\mathfrak{A}$ is commutative, the norm (1.25) on $\mathfrak{A}$ simply becomes $\|A + iB\| = \|A^2 + B^2\|^{\frac{1}{2}}$, where $A, B \in \mathfrak{A}_{\mathbb{R}}$. All axioms for a norm are then easily derived from (1.9) and (1.10).

The proof that (1.25) is a norm also in the noncommutative case, as well as the proof of (1.14), will be given at the end of 1.4, too. □

One could replace the minus sign on the right-hand side of (1.23) by a plus sign; that choice leads to a C^*-algebra as well, which is anti-isomorphic to the one based on the minus sign.

1.2 Spectrum and Commutative C^*-Algebras

We are going to examine to what extent the closely related notions of spectrum and functional calculus of a (bounded) self-adjoint operator A on a Hilbert space $\mathcal{H}$ generalize to the context of C^*-algebras and JLB-algebras. On the way we discuss the structure of commutative C^*-algebras. In this section we do not use Theorem 1.1.9, except in the commutative case, for the outstanding part of the proof of this theorem will depend on some of the results below.

Recall that the spectrum $\sigma(A)$ of $A \in \mathfrak{B}(\mathcal{H})$ is the set of those $z \in \mathbb{C}$ for which $A - z\mathbb{I}$ has no (bounded two-sided) inverse; when A is self-adjoint, the **spectral radius** $r(A)$ appears in the fundamental equality

$$\|A\| = r(A) := \sup\{|z| \mid z \in \sigma(A)\}. \tag{1.26}$$

Since the presence of a unit is crucial in these definitions, we have to go through the following considerations. A **unit** $\mathbb{I}$ in a JB-algebra $\mathfrak{A}_{\mathbb{R}}$ is an element such that $A \circ I = A$ for all $A \in \mathfrak{A}_{\mathbb{R}}$; a JB-algebra with a unit is called **unital.** When $\mathfrak{A}_{\mathbb{R}}$ is a JLB-algebra, its unit becomes a unit of the C^*-algebra $\mathfrak{A}$, in that $\mathbb{I}A = A\mathbb{I} = A$ for all $A \in \mathfrak{A}$. This follows by putting $B = C = \mathbb{I}$ in (1.4), implying $\{A, \mathbb{I}\} = 0$ for all A, and subsequently applying (1.23). In any case, taking the adjoint of $\mathbb{I}^*\mathbb{I} = \mathbb{I}^*$ yields $\mathbb{I}^*\mathbb{I} = \mathbb{I}$; hence $\mathbb{I}^* = \mathbb{I}$. Also, (1.15) then implies $\|\mathbb{I}\| = 1$.

When a C^*-algebra or a JLB-algebra has no unit, one may add one.

Proposition 1.2.1. *For every C^*-algebra without unit there exists a unique unital C^*-algebra $\mathfrak{A}_{\mathbb{I}}$, called the* **unitization** *of $\mathfrak{A}$, and an isometric (hence injective) morphism $\mathfrak{A} \to \mathfrak{A}_{\mathbb{I}}$, such that $\mathfrak{A}_{\mathbb{I}}/\mathfrak{A} \simeq \mathbb{C}$.*

Let $\mathfrak{B}(\mathfrak{A})$ be the Banach algebra of all bounded linear maps on $\mathfrak{A}$. Whether or not $\mathfrak{A}$ is unital, the map $\rho : \mathfrak{A} \to \mathfrak{B}(\mathfrak{A})$, given by

$$\rho(A)B := AB, \tag{1.27}$$

is isometric. To see this, note that $\|\rho(A)\| \leq \|A\|$ by (1.14), whereas the opposite inequality follows from (1.15).

Now let $\mathfrak{A}$ be a nonunital C^*-algebra, and form $\mathfrak{A}_{\mathbb{I}} := \mathfrak{A} \oplus \mathbb{C}$. Extend ρ to $\mathfrak{A}_{\mathbb{I}}$ by $\rho(A\dot{+}z)B := AB + zB$, so that $\rho(0\dot{+}1) = \mathbb{I}$ (the unit in $\mathfrak{B}(\mathfrak{A})$). Equipped with the norm (1.17) and the algebraic structure of $\mathfrak{B}(\mathfrak{A})$, and with the involution $\rho(A\dot{+}z)^* := \rho(A^*) + \overline{z}\mathbb{I}$, the vector space $\rho(\mathfrak{A}_{\mathbb{I}})$ is easily shown to be a unital C^*-algebra. Since ρ is a vector space isomorphism between $\mathfrak{A}_{\mathbb{I}}$ and $\rho(\mathfrak{A}_{\mathbb{I}})$, one may transfer the C^*-algebraic structure on the latter to $\mathfrak{A}_{\mathbb{I}}$. Restricted to $\mathfrak{A} \subset \mathfrak{A}_{\mathbb{I}}$, one recovers $\mathfrak{A}$ as a C^*-algebra. Uniqueness follows from 1.2.4.4 below. □

Definition 1.2.2. *Let $\mathfrak{A}$ be either a unital C^*-algebra, or the complexification of a unital JLB-algebra $\mathfrak{A}_{\mathbb{R}}$. The* **spectrum** *$\sigma(A)$ of $A \in \mathfrak{A}$ is defined as the set of those $z \in \mathbb{C}$ for which $A - z\mathbb{I}$ has no (two-sided) inverse in $\mathfrak{A}$.*

When $\mathfrak{A}$ is nonunital, one puts $\sigma(A) := \sigma_{\mathbb{I}}(A\dot{+}0)$, where $\sigma_{\mathbb{I}}$ stands for the spectrum in the unitization of $\mathfrak{A}$.

In the nonunital case 0 always lies in $\sigma(A)$, as it is obvious from 1.2.1 that $A \in \mathfrak{A}$ never has an inverse in $\mathfrak{A}_{\mathbb{I}}$. The theory of Banach algebras shows that $\sigma(A)$ is a compact subset of $\mathbb{C}$. For later use we note that

$$\sigma(zA) = z\sigma(A); \tag{1.28}$$

$$\sigma(AB) \cup \{0\} = \sigma(BA) \cup \{0\} \tag{1.29}$$

for all $A, B \in \mathfrak{A}$; the first property is obvious, and the second follows, because for $z \neq 0$ the invertibility of $AB - z$ implies the invertibility of $BA - z$. Namely, one computes that $(BA - z)^{-1} = B(AB - z)^{-1}A - z^{-1}\mathbb{I}$.

For any locally compact Hausdorff space X, we regard the space $C_0(X)$ of all continuous functions on X that vanish at infinity as a Banach space in the sup-norm. A basic fact of topology and analysis is that $C_0(X)$ is complete in this norm. Convergence in the sup-norm is the same as uniform convergence. What's more, it is easily verified that $C_0(X)$ is a commutative C^*-algebra under pointwise addition, multiplication, and complex conjugation (defining the involution). When X is compact, the function 1_X, which is 1 for every x, is the unit $\mathbb{I}$. One checks that the spectrum of $f \in C(X)$ is simply the set of values of f. On $C_0(X)$, with X noncompact, one has to supplement this set with zero.

Theorem 1.2.3. *Let $\mathfrak{A}$ be a commutative C^*-algebra. There exists a locally compact Hausdorff space X for which $\mathfrak{A}$ is isomorphic to $C_0(X)$. When $\mathfrak{A}$ is unital, X is compact, so that $\mathfrak{A} \simeq C(X)$. The space X is unique up to homeomorphism.*

Similarly, an associative JLB-algebra $\mathfrak{A}_{\mathbb{R}}$ is isomorphic to some $C_0(X, \mathbb{R})$, where X is locally compact; when $\mathfrak{A}_{\mathbb{R}}$ has a unit, X is compact.

For simplicity we assume that $\mathfrak{A}$ is unital; if it isn't, one would start by adjoining a unit. The proof is based on a technique that applies to general commutative unital

Banach algebras; we state the main facts without proof. Consider the space $\Delta(\mathfrak{A})$ of all nonzero multiplicative linear functionals ω on $\mathfrak{A}$ (that is, $\omega : \mathfrak{A} \to \mathbb{C}$ satisfies $\omega(AB) = \omega(A)\omega(B)$ for all A, B). Each $\omega \in \Delta(\mathfrak{A})$ is continuous, and satisfies $\|\omega\| = \omega(\mathbb{I}) = 1$. Thence it is easily seen that $\Delta(\mathfrak{A})$ is a closed subspace of $\mathfrak{A}^*$ with the w^*-topology. By the Banach–Alaoglu theorem, $\Delta(\mathfrak{A})$ is therefore a compact Hausdorff space in the relative w^*-topology.

The **Gelfand transform** of $A \in \mathfrak{A}$ is the function $\hat{A}$ on $\Delta(\mathfrak{A})$ defined by

$$\hat{A}(\omega) := \omega(A). \tag{1.30}$$

Since the relative w^*-topology on $\Delta(\mathfrak{A})$ coincides with the weakest topology that makes all functions $\hat{A}$ continuous, it is clear that the Gelfand transform maps $\mathfrak{A}$ into $C(\Delta(\mathfrak{A}))$. It is immediate that the image of $\mathfrak{A}$ in $C(\Delta(\mathfrak{A}))$ separates points. Regarding $C(\Delta(\mathfrak{A}))$ as a commutative Banach algebra in the sup-norm, as explained above, the multiplicativity of each $\omega \in \Delta(\mathfrak{A})$ implies that the Gelfand transform is a homomorphism. The spectrum of $A \in \mathfrak{A}$ coincides with the set of values of $\hat{A}$ on $\Delta(\mathfrak{A})$; in other words,

$$\sigma(A) = \sigma(\hat{A}) = \{\hat{A}(\omega) \mid \omega \in \Delta(\mathfrak{A})\}. \tag{1.31}$$

This implies that

$$\|\hat{A}\|_\infty = r(A), \tag{1.32}$$

where the spectral radius $r(A)$ is defined in (1.26). In any Banach algebra, commutative or not, one has

$$r(A) = \lim_{n\to\infty} \|A^n\|^{1/n}. \tag{1.33}$$

Now assume that $\mathfrak{A}$ is a commutative C^*-algebra; accordingly, regard $C(\Delta(\mathfrak{A}))$ as a commutative C^*-algebra. We first show that $\omega \in \Delta(\mathfrak{A})$ is real on $\mathfrak{A}_{\mathbb{R}}$. Pick $A \in \mathfrak{A}_{\mathbb{R}}$, and suppose that $\omega(A) = \alpha + i\beta$, where $\alpha, \beta \in \mathbb{R}$. Since $\omega(\mathbb{I}) = 1$, one has $\omega(B) = i\beta$, where $B := A - \alpha\mathbb{I}$ is self-adjoint. Hence for $t \in \mathbb{R}$ one computes $|\omega(B + it\mathbb{I})|^2 = \beta^2 + 2t\beta + t^2$. On the other hand, using $\|\omega\| = 1$ and (1.15) we estimate $|\omega(B + it\mathbb{I})|^2 \leq \|B\|^2 + t^2$. Hence $\beta^2 + t\beta \leq \|B\|^2$ for all $t \in \mathbb{R}$. For $\beta > 0$ this is impossible. For $\beta < 0$ we repeat the argument with B replaced by $-B$. Hence $\beta = 0$, so that $\omega(A)$ is real when $A = A^*$. Consequently, by (1.30) the function $\hat{A}$ is real-valued. Writing $\varphi(A) := \hat{A}$, condition (1.20) follows.

Secondly, for $A \in \mathfrak{A}_{\mathbb{R}}$ one combines (1.15) with (1.33) to obtain $\|A\| = r(A)$, which with (1.33) implies $\|\hat{A}\|_\infty = \|A\|$. For general A this equality then follows via (1.15) in both $\mathfrak{A}$ and $C(\Delta(\mathfrak{A}))$, and the fact that $A^*A \in \mathfrak{A}_{\mathbb{R}}$. Thus the Gelfand transform is isometric, and therefore injective. Finally, surjectivity follows from the Stone–Weierstrass theorem.

The uniqueness of X follows from the fact that when X and Y are compact Hausdorff spaces, the commutative Banach algebras $C(X)$ and $C(Y)$ are isomorphic iff X and Y are homeomorphic; this is equivalent to the statement that $\Delta(C(X))$ is homeomorphic to X. The homeomorphism is given by letting $x \in X$ correspond with $\omega_x \in \Delta(C(X))$, defined by $\omega_x(f) := f(x)$. The assumption that X is compact and Hausdorff, hence normal, is used to prove that the evaluation map $x \mapsto \omega_x$

is injective, whereas the compactness of X implies that the evaluation map is surjective.

The case of a commutative JLB-algebra $\mathfrak{A}_{\mathbb{R}}$ may be reduced to that of a commutative C^*-algebra by complexification; see Theorem 1.1.9. □

Theorem 1.2.4. *Let $\mathfrak{A}$ be either a unital C^*-algebra or the complexification of a unital JLB-algebra $\mathfrak{A}_{\mathbb{R}}$.*

1. *The spectrum $\sigma(A)$ of $A \in \mathfrak{A}_{\mathbb{R}}$ in $\mathfrak{A}$ is equal to its spectrum in the C^*-algebra $C^*(A)$ generated by A and $\mathbb{I}$. In particular, $\sigma(A)$ is a subset of the real line.*
2. *The compact Hausdorff space $\Delta(C^*(A))$ of Theorem 1.2.3 is homeomorphic to $\sigma(A)$, and the C^*-algebra $C^*(A)$ of the preceding item is isomorphic to $C(\sigma(A))$. Under this isomorphism the function $\hat{A} \in C(X)$ is mapped into $\mathrm{id}_{\sigma(A)} : t \mapsto t$.*
3. *The* **continuous functional calculus** *for self-adjoint operators A on a Hilbert space applies verbatim to $\mathfrak{A}_{\mathbb{R}}$: In particular, for each $A \in \mathfrak{A}_{\mathbb{R}}$ and $f \in C(\sigma(A))$ there exists an operator $f(A) \in \mathfrak{A}$ that is the obvious expression when f is polynomial (and in the general case is given by uniformly approximating f by polynomials on the basis of the Stone–Weierstrass theorem), and has the properties*

$$\sigma(f(A)) = f(\sigma(A)); \tag{1.34}$$
$$\|f(A)\| = \|f\|_\infty. \tag{1.35}$$

4. *For $A \in \mathfrak{A}_{\mathbb{R}}$ the norm is given by (1.26); for general A one has $\|A\| = \sqrt{r(A^*A)}$. Hence the norm in a C^*-algebra is unique, in that a *-algebra that is a C^*-algebra in some norm admits no other norm in which it is a C^*-algebra.*

If $A = A^*$, then $A_z := A - z$ is normal for any $z \in \mathbb{C}$ (i.e., A_z commutes with its adjoint $A_{\bar{z}}$). Suppose that $z \notin \sigma(A)$, so that A_z is invertible. Consider the commutative C^*-algebra $C^*(A_z, A_z^{-1})$ generated by A_z, its inverse, and the unit. By Theorem 1.2.3 one has $C^*(A_z, A_z^{-1}) \simeq C(X)$, where $X = \Delta(C^*(A_z, A_z^{-1}))$. Since A_z is invertible in $C^*(A_z, A_z^{-1})$, it must be that $\hat{A}_z(x) \neq 0$ for all $x \in X$. It is then elementary that $\hat{A}_z^{-1}$ is a uniform limit of polynomials in A_z, $\hat{A}_{\bar{z}}$, and 1_X. Transferring this back to $C^*(A_z, A_z^{-1})$ by the inverse of the Gelfand transform, it follows that $C^*(A_z, A_z^{-1}) = C^*(A_z) = C^*(A)$. Hence when $A - z$ is invertible in $\mathfrak{A}$ its inverse lies in $C^*(A)$, which implies the first claim in 1.2.4.1.

Consider 1.2.3 and its proof with $\mathfrak{A} = C^*(A)$. We see from (1.31) and the fact that $\hat{A}$ is real-valued for $A \in \mathfrak{A}_{\mathbb{R}}$ that the spectrum of A in $C^*(A)$ is real. When now $\mathfrak{A}$ has the meaning of the present Theorem 1.2.4, the second claim in 1.2.4.1 follows from the first one just proved.

For 1.2.4.2, consider the map $\hat{A} : X \to \mathbb{R}$. It is clear from (1.31) that the image of $\hat{A}$ is $\sigma(A)$. To prove injectivity, assume $\hat{A}(\omega_1) = \hat{A}(\omega_2)$. Then $\omega_1(A) = \omega_2(A)$ by (1.30), whence $\omega_1(A)^n = \omega_2(A)^n$ by multiplicativity of $\omega_i \in \Delta(C^*(A))$. Since the linear span of all polynomials is dense in $C^*(A)$, and the ω_i are continuous, this yields $\omega_1 = \omega_2$. The map $\hat{A}$ is continuous, because $\hat{A} \in C(X)$ by 1.2.3. To prove continuity of the inverse, one checks that for $z \in \sigma(A)$ the functional $\hat{A}^{-1}(z) \in \Delta(C^*(A))$ maps A to z (and hence A^n to z^n, etc.). Finally, given the homeomorphism $\Delta(C^*(A)) \simeq \sigma(A)$, the second isomorphism in $C^*(A) \simeq C(\Delta(C^*(A))) \simeq C(\sigma(A))$ follows from the topological fact that a compact Hausdorff space X is determined by $C(X)$, and vice versa; cf. the proof of 1.2.3.

The existence of the continuous functional calculus should now be obvious. Since $f(\sigma(A))$ is the set of values of f on $\sigma(A)$, (1.34) follows from (1.31), with A replaced by $f(A)$. The fact that for C^*-algebras the Gelfand transform is isometric yields (1.35).

The corresponding statements for a JLB-algebra follow by complexification, using the commutative part of 1.1.9.

The first claim in 1.2.4.4 follows from (1.35) with $f = \text{id}$. The second claim follows from the first, (1.15), and the property $A^*A \in \mathfrak{A}_{\mathbb{R}}$. Hence the norm is determined by the algebraic structure. ∎

For later use we record that for $A \in \mathfrak{A}_{\mathbb{R}}$, Theorem 1.2.4 implies

$$\sigma(A) = 0 \Leftrightarrow A = 0. \tag{1.36}$$

1.3 Positivity, Order, and Morphisms

Recall that a (bounded) operator $A \in \mathfrak{B}(\mathcal{H})$ on a Hilbert space is called **positive** when $(\Psi, A\Psi) \geq 0$ for all $\Psi \in \mathcal{H}$; this property is equivalent to $A^* = A$ and $\sigma(A) \subseteq \mathbb{R}^+$. This notion of positivity induces a partial ordering $\leq$ in $\mathfrak{B}(\mathcal{H})$, in which $A \leq B$ when $B - A \geq 0$. Our aim is to generalize these concepts to C^*-algebras and JLB-algebras.

Definition 1.3.1. *A* **partially ordered vector space** $(\mathfrak{A}_{\mathbb{R}}, \leq)$ *consists of a real vector space* $\mathfrak{A}_{\mathbb{R}}$ *and either one of the following equivalent data:*

- *A* **positive cone** $\mathfrak{A}^+$ *in* $\mathfrak{A}_{\mathbb{R}}$*; this is a subset for which* (i) $A \in \mathfrak{A}^+$ *and* $t \in \mathbb{R}^+$ *implies* $tA \in \mathfrak{A}^+$, (ii) $A, B \in \mathfrak{A}^+$ *implies* $A+B \in \mathfrak{A}^+$, *and* (iii) $\mathfrak{A}^+ \cap -\mathfrak{A}^+ = 0$.
- *A* **linear partial ordering**, *i.e., a partial ordering* $\leq$ *in which* $A \leq B$ *implies* $A + C \leq B + C$ *for all* $C \in \mathfrak{A}_{\mathbb{R}}$ *and* $tA \leq tB$ *for all* $t \in \mathbb{R}^+$.

The equivalence between these two structures is as follows: Given $\mathfrak{A}^+_{\mathbb{R}}$ one defines $A \leq B$ if $B - A \in \mathfrak{A}^+_{\mathbb{R}}$, and given $\leq$ one puts $\mathfrak{A}^+_{\mathbb{R}} = \{A \in \mathfrak{A}_{\mathbb{R}} \mid 0 \leq A\}$.

Definition 1.3.2. *Let* $\mathfrak{A}_{\mathbb{R}}$ *be a* JLB*-algebra or the self-adjoint part of a* C^**-algebra. An element* $A \in \mathfrak{A}_{\mathbb{R}}$ *is called* **positive** *when its spectrum is positive; i.e.,* $\sigma(A) \subset \mathbb{R}^+$. *We write* $A \geq 0$ *or* $A \in \mathfrak{A}^+$, *where*

$$\mathfrak{A}^+ := \{A \in \mathfrak{A}_{\mathbb{R}} | \sigma(A) \subset \mathbb{R}^+\}. \tag{1.37}$$

It is immediate from (1.31) that $A \in \mathfrak{A}_{\mathbb{R}}$ is positive iff its Gelfand transform $\hat{A}$ is pointwise positive in $C(\sigma(A))$.

Theorem 1.3.3. *The set (1.37) of all positive elements of a C^*-algebra $\mathfrak{A}$ or a JLB-algebra $\mathfrak{A}_{\mathbb{R}}$ is a positive cone. This cone may alternatively be expressed as*

$$\mathfrak{A}^+_{\mathbb{R}} = \{A^2 \mid A \in \mathfrak{A}_{\mathbb{R}}\} \tag{1.38}$$

$$= \{B^*B \mid B \in \mathfrak{A}\}. \tag{1.39}$$

Property (i) in 1.3.1 follows from (1.28). Since $\sigma(A) \subseteq [0, r(A)]$, we have $|c - t| \leq c$ for all $t \in \sigma(A)$ and all $c \geq r(A)$. Hence $\sup_{t\in\sigma(A)} |c1_{\sigma(A)} - \hat{A}| \leq c$ by (1.31) and 1.2.4.2, so that $\|c1_{\sigma(A)} - \hat{A}\|_\infty \leq c$. Gelfand transforming back to $C^*(A)$, this implies $\|c\mathbb{I} - A\| \leq c$ for all $c \geq \|A\|$ by 1.2.4.3. Inverting this argument, one sees that if $\|c\mathbb{I} - A\| \leq c$ for some $c \geq \|A\|$, then $\sigma(A) \subset \mathbb{R}^+$. Using this with A replaced by $A + B$ and $c = \|A\| + \|B\|$ leads to property (ii). Finally, when $A \in \mathfrak{A}^+$ and $A \in -\mathfrak{A}^+$, it must be that $\sigma(A) = 0$, hence $A = 0$ by (1.36). This proves property (iii).

If $\sigma(A) \subset \mathbb{R}^+$ and $A = A^*$, then $\sqrt{A} \in \mathfrak{A}_{\mathbb{R}}$ is defined by the continuous functional calculus for $f = \sqrt{\cdot}$ and satisfies $\sqrt{A}^2 = A$. Hence $\mathfrak{A}^+ \subseteq \{A^2 \mid A \in \mathfrak{A}_{\mathbb{R}}\}$. The opposite inclusion follows from (1.34) and 1.2.4.2. This proves (1.38).

The inclusion $\mathfrak{A}^+ \subseteq \{B^*B \mid B \in \mathfrak{A}\}$ is trivial from (1.38).

Lemma 1.3.4. *Every $A \in \mathfrak{A}_{\mathbb{R}}$ has a decomposition $A = A_+ - A_-$, where $A_+, A_- \in \mathfrak{A}^+$ and $A_+A_- = 0$. Moreover, $\|A_\pm\| \leq \|A\|$.*

Apply the continuous functional calculus with $f = \mathrm{id}_{\sigma(A)} = f_+ - f_-$, where $\mathrm{id}_{\sigma(A)}(t) = t$, $f_+(t) = \max\{t, 0\}$, and $f_-(t) = \max\{-t, 0\}$. The bound follows from (1.35) with A replaced by $A_\pm$. ■

Apply this decomposition to $A = B^*B$ (noting that $A = A^*$); it follows that $(A_-)^3 = -(BA_-)^*BA_-$. Since $\sigma(A_-) \subset \mathbb{R}^+$ as A_- is positive, we see from (1.34) with $f(t) = t^3$ that $(A_-)^3 \geq 0$. Hence $-(BA_-)^*BA_- \geq 0$.

Lemma 1.3.5. *If $-C^*C \in \mathfrak{A}^+$ for some $C \in \mathfrak{A}$, then $C = 0$.*

Write $C = D + iE$, where $D, E \in \mathfrak{A}_{\mathbb{R}}$ (cf. (1.21)), so that $C^*C = 2D^2 + 2E^2 - CC^*$. Applying (1.29) with A replaced by C and B by C^*, we see that the assumption $\sigma(C^*C) \subset \mathbb{R}^-$ implies $\sigma(CC^*) \subset \mathbb{R}^-$; since C^*C is the sum of three positive terms, and $\mathfrak{A}^+$ is a positive cone, it follows that $C^*C \in \mathfrak{A}^+$. Hence the starting assumption $\sigma(C^*C) \subset \mathbb{R}^-$ implies $\sigma(C^*C) \subset \mathbb{R}^+$, so that $\sigma(C^*C) = 0$. Hence $C^*C = 0$ by (1.36).

In a C^*-algebra this implies $C = 0$ by (1.15). In a complexified JLB-algebra we replace C by C^* in the above argument, so that $CC^* = 0$ as well as $C^*C = 0$; hence $D^2 + E^2 = 0$, whence $D = E = 0$ by (1.8). ■

The last claim before the lemma therefore implies $BA_- = 0$. As $(A_-)^3 = -(BA_-)^*BA_- = 0$, we see that $(A_-)^3 = 0$, and finally $A_- = 0$ by the continuous functional calculus with $f(t) = t^{1/3}$. Hence $B^*B = A_+$, which lies in $\mathfrak{A}^+$. ■

When $A = A^*$ one checks the validity of

$$-\|A\|\,\mathbb{I} \leq A \leq \|A\|\,\mathbb{I} \tag{1.40}$$

by taking the Gelfand transform of $C^*(A)$. The implication

$$-B \le A \le B \implies \|A\| \le \|B\| \tag{1.41}$$

then follows, because $-B \le A \le B$ and (1.40) with A replaced by B yield $-\|B\|\,\mathbb{I} \le A \le \|B\|\,\mathbb{I}$, so that $\sigma(A) \subseteq [-\|B\|, \|B\|]$; hence $\|A\| \le \|B\|$ by (1.26).

An important consequence of (1.39) is the fact that inequalities of the type $A_1 \le A_2$ for $A_1, A_2 \in \mathfrak{A}_{\mathbb{R}}$ are stable under conjugation by arbitrary elements $B \in \mathfrak{A}$, so that $A_1 \le A_2$ implies $B^*A_1B \le B^*A_2B$. This is because $A_1 \le A_2$ is the same as $A_2 - A_1 \ge 0$; by (1.39) there is an $A_3 \in \mathfrak{A}$ such that $A_2 - A_1 = A_3^*A_3$. But then $(A_3B)^*A_3B \ge 0$, and this is nothing but $B^*A_1B \le B^*A_2B$. For example, replace A in (1.40) by A^*A, and use (1.15) in a C^*-algebra, or (1.25) in a complexified JLB-algebra. This yields $A^*A \le \|A\|^2\mathbb{I}$. Applying the above principle to any $A, B \in \mathfrak{A}$ gives

$$B^*A^*AB \le \|A\|^2B^*B. \tag{1.42}$$

Definition 1.3.6. *A* **positive map** $Q : \mathfrak{A} \to \mathfrak{B}$ *between two C^*-algebras is a linear map with the property that $A \ge 0$ in $\mathfrak{A}$ implies $Q(A) \ge 0$ in $\mathfrak{B}$.*

Proposition 1.3.7. *A positive map between C^*-algebras is $*$-preserving and bounded.*

One infers from 1.3.4 that $Q(\mathfrak{A}_{\mathbb{R}}) \subseteq \mathfrak{B}_{\mathbb{R}}$; the $\mathbb{C}$-linearity of Q then proves the first claim.

For the second claim, let us first show that boundedness on $\mathfrak{A}^+$ implies boundedness on $\mathfrak{A}$. Using (1.21) and 1.3.4, we can write $A = A'_+ - A'_- + iA''_+ - iA''_-$, where A'_+ etc. are positive. Since $\|A'\| \le \|A\|$ and $\|A''\| \le \|A\|$ by (1.21), we have $\|B\| \le \|A\|$ for $B = A'_+, A'_-, A''_+$, or A''_- by 1.3.4. Hence if $\|Q(B)\| \le c\|B\|$ for all $B \in \mathfrak{A}^+$ and some $c > 0$, then $\|Q(A)\| \le 4c\|A\|$.

Now assume that Q is not bounded; by the previous argument it is not bounded on $\mathfrak{A}^+$, so that for each $n \in \mathbb{N}$ there is an $A_n \in \mathfrak{A}_1^+$ such that $\|Q(A_n)\| \ge n^3$ (here $\mathfrak{A}_1^+$ consists of all $A \in \mathfrak{A}^+$ with $\|A\| \le 1$). The series $\sum_{n=0}^{\infty} n^{-2}A_n$ obviously converges to some $A \in \mathfrak{A}^+$. Since Q is positive, we have $Q(A) \ge n^{-2}Q(A_n) \ge 0$ for each n. Hence by (1.41), $\|Q(A)\| \ge n^{-2}\|Q(A_n)\| \ge n$ for all $n \in \mathbb{N}$, which is impossible. Thus Q is bounded on $\mathfrak{A}^+$, and therefore on $\mathfrak{A}$ by the previous paragraph. ■

Since a morphism $\varphi : \mathfrak{A} \to \mathfrak{B}$ satisfies $\varphi(B^*B) = \varphi(B)^*\varphi(B)$, it is clear from (1.39) that a morphism is a positive map. For later reference we collect this, and other good properties of morphisms. In preparation, we define a **left ideal** in a C^*-algebra $\mathfrak{A}$ as a closed linear subspace $\mathfrak{J} \subseteq \mathfrak{A}$ such that $A \in \mathfrak{J}$ implies $BA \in \mathfrak{J}$ for all $B \in \mathfrak{A}$. Similarly, a **right ideal** is a closed linear subspace $\mathfrak{J} \subseteq \mathfrak{A}$ such that $A \in \mathfrak{J}$ implies $AB \in \mathfrak{J}$ for all $B \in \mathfrak{A}$. An **ideal** is both a left and a right ideal.

A proper ideal cannot contain a unit $\mathbb{I}$; in order to prove properties of ideals one needs a suitable replacement of a unit.

Definition 1.3.8. *An* **approximate unit** *in a nonunital C^*-algebra $\mathfrak{A}$ consists of a directed set Λ (i.e., a set with a partial order $\leq$ in which for all λ_1, λ_2 there is a $\lambda \geq \lambda_i$, $i = 1, 2$) and a family $\{\mathbb{I}_\lambda\}_{\lambda \in \Lambda}$ of elements of $\mathfrak{A}$ for which $\mathbb{I}_\lambda^* = \mathbb{I}_\lambda$ and $\sigma(\mathbb{I}_\lambda) \subset [0, 1]$ (whence $\|\mathbb{I}_\lambda\| \leq 1$), so that for all $A \in \mathfrak{A}$ one has*

$$\lim_{\lambda \to \infty} \|\mathbb{I}_\lambda A - A\| = \lim_{\lambda \to \infty} \|A\mathbb{I}_\lambda - A\| = 0. \tag{1.43}$$

For example, an approximate unit in $C_0(\mathbb{R})$ may be constructed with $\Lambda = \mathbb{N}$, taking $\mathbb{I}_n$ to be a continuous function that is 1 on $[-n, n]$ and vanishes for $|x| > n + 1$. More generally, it can be shown that every nonunital C^*-algebra $\mathfrak{A}$ has an approximate unit; when $\mathfrak{A}$ is separable, Λ may be chosen countable. The technique of approximate units allows us to prove the main properties of ideals in C^*-algebras.

Theorem 1.3.9. *Let $\mathfrak{J}$ be an ideal in a C^*-algebra $\mathfrak{A}$.*

1. *If $J \in \mathfrak{J}$ then $J^* \in \mathfrak{J}$; in other words, every ideal in a C^*-algebra is self-adjoint.*
2. *The quotient $\mathfrak{A}/\mathfrak{J}$ is a C^*-algebra in the norm $\|\tau(A)\| := \inf_{J \in \mathfrak{J}} \|A + J\|$, the multiplication $\tau(A)\tau(B) := \tau(AB)$, and the involution $\tau(A)^* := \tau(A^*)$.*

Note that the involution in $\mathfrak{A}/\mathfrak{J}$ is well-defined because of 1.3.9.1.

Put $\mathfrak{J}^* := \{A^* | A \in \mathfrak{J}\}$, and note that $\mathfrak{J} \cap \mathfrak{J}^*$ is a C^*-subalgebra of $\mathfrak{A}$. Hence it has an approximate unit $\{\mathbb{I}_\lambda\}$. Pick $J \in \mathfrak{J}$, and use (1.15) and 1.3.8 to estimate

$$\|J^* - J^*\mathbb{I}_\lambda\|^2 \leq \|(J^*J - J^*J\mathbb{I}_\lambda)\| + \|(JJ^* - JJ^*\mathbb{I}_\lambda)\|.$$

Now J^*J and JJ^* both lie in $\mathfrak{J} \cap \mathfrak{J}^*$, so that both terms on the right-hand side vanish for $\lambda \to \infty$. Hence J^* is a norm-limit of elements in $\mathfrak{J}$; since $\mathfrak{J}$ is closed, it follows that $J^* \in \mathfrak{J}$.

We omit the well-known proof that $\mathfrak{A}/\mathfrak{J}$ is a Banach algebra in the given norm and multiplication. To prove the C^*-property (1.15), we first note that

$$\|\tau(A)\| = \lim_{\lambda \to \infty} \|A - A\mathbb{I}_\lambda\|, \tag{1.44}$$

for any $A \in \mathfrak{A}$ and approximate unit $\{\mathbb{I}_\lambda\}$ in $\mathfrak{J}$. To prove this, we first add a unit to $\mathfrak{A}$ if necessary. For any $J \in \mathfrak{J}$ we have $A - A\mathbb{I}_\lambda = (A + J)(\mathbb{I} - \mathbb{I}_\lambda) + J(\mathbb{I}_\lambda - \mathbb{I})$, so that $\|A - A\mathbb{I}_\lambda\| \leq \|A + J\| \, \|\mathbb{I} - \mathbb{I}_\lambda\| + \|J\mathbb{I}_\lambda - J\|$. Since

$$\|\mathbb{I} - \mathbb{I}_\lambda\| \leq 1 \tag{1.45}$$

from 1.3.8 and the proof of 1.3.3, we obtain $\lim_{\lambda \to \infty} \|A - A\mathbb{I}_\lambda\| \leq \|A + J\|$. For each $\epsilon > 0$ we can choose $J \in \mathfrak{J}$ such that $\|\tau(A)\| + \epsilon \geq \|A + J\|$. Using this J in the previous inequality, letting $\epsilon \to 0$, and noting the obvious $\|A - A\mathbb{I}_\lambda\| \geq \|\tau(A)\|$, we obtain (1.44).

Successively using (1.44), (1.15) in $\mathfrak{A}$, (1.45), (1.44) once again, and the definition of the C^*-operations in $\mathfrak{A}/\mathfrak{J}$, we obtain $\|\tau(A)\|^2 \leq \|\tau(A)\tau(A)^*\|$. By the argument preceding 1.1.7, this implies (1.15). □

Theorem 1.3.10. *Let $\varphi : \mathfrak{A} \to \mathfrak{B}$ be a morphism between C^*-algebras.*

1. *The kernel of φ is an ideal in $\mathfrak{A}$. Conversely, every ideal in a C^*-algebra is the kernel of some morphism.*

2. *One has* $\|\varphi\| = 1$, *and therefore* $\|\varphi(A)\| \leq \|A\|$ *for all* $A \in \mathfrak{A}$.
3. *The map* φ *is isometric when it is injective.*
4. *The image* $\varphi(\mathfrak{A})$ *is a* C^**-subalgebra of* $\mathfrak{B}$.
5. *The map* φ *is positive.*

It is clear that $\sigma(\varphi(A^*A)) \subseteq \sigma(A^*A)$, so that the inequality in 1.3.10.2 follows from 1.2.4.4. It follows that φ is continuous, so that $\ker(\varphi)$ is closed. It is then obvious from (1.19) that $\ker(\varphi)$ is an ideal. On the other hand, a given ideal $\mathfrak{J}$ is the kernel of the canonical projection $\tau : \mathfrak{A} \to \mathfrak{A}/\mathfrak{J}$. Now $\mathfrak{A}/\mathfrak{J}$ is a C^*-algebra and $\varphi := \tau$ is a morphism with $\mathfrak{J} = \ker(\varphi)$.

Assume that there is a $B \in \mathfrak{A}$ for which $\|\varphi(B)\| \neq \|B\|$. By (1.15), (1.19), and (1.20) this implies $\|\varphi(B^*B)\| \neq \|B^*B\|$. Put $A := B^*B$, noting that $A^* = A$. By (1.26) we must have $\sigma(A) \neq \sigma(\varphi(A))$. Since $\sigma(\varphi(A)) \subseteq \sigma(A)$ in any case, this implies $\sigma(\varphi(A)) \subset \sigma(A)$. By Urysohn's lemma there is a nonzero $f \in C(\sigma(A))$ that vanishes on $\sigma(\varphi(A))$, so that $f(\varphi(A)) = 0$. By the continuous functional calculus we have $\varphi(f(A)) = 0$, proving 1.3.10.3 by reductio ad absurdum.

Define $\psi : \mathfrak{A}/\ker(\varphi) \to \mathfrak{B}$ by $\psi \circ \tau = \varphi$, with $\tau : \mathfrak{A} \to \mathfrak{A}/\ker(\varphi)$ the canonical projection. Then ψ is an injective morphism, so that it is isometric by 1.3.10.3. Hence $\|\varphi\| = 1$, since $\|\tau\| = 1$. Since $\psi(\mathfrak{A}/\ker(\varphi)) = \varphi(\mathfrak{A})$, it follows that φ has closed range in $\mathfrak{B}$. Since φ is a morphism, this implies that $\varphi(\mathfrak{A})$ is a C^*-algebra in the norm of $\mathfrak{B}$. ∎

1.4 States

We now change our perspective, and pass from observables to states.

Definition 1.4.1. *A* **state** *on a* C^**-algebra* $\mathfrak{A}$ *is a linear map* $\omega : \mathfrak{A} \to \mathbb{C}$ *for which* $\omega(A) \geq 0$ *for all* $A \in \mathfrak{A}^+_{\mathbb{R}}$ (**positivity**) *and* $\|\omega\| = 1$ (**normalization**). *The* **state space** $\mathcal{S}(\mathfrak{A})$ *of* $\mathfrak{A}$ *is the set of all states on* $\mathfrak{A}$.

For example, on $\mathfrak{A} = \mathfrak{B}(\mathcal{H})$ every unit vector $\Omega \in \mathcal{H}$ defines a state ω by

$$\omega(A) = (\Omega, \pi(A)\Omega). \tag{1.46}$$

This is, indeed, the original notion of a state as used in quantum mechanics.

Combining 1.3.4 with positivity, we see that a state is real-valued on $\mathfrak{A}_{\mathbb{R}}$; in view of (1.21) we then infer that a state is a **Hermitian functional** on $\mathfrak{A}$, in that

$$\omega(A^*) = \overline{\omega(A)} \tag{1.47}$$

for all $A \in \mathfrak{A}$. In particular, a state is determined by its values on $\mathfrak{A}^+$. Combining the positivity of ω with (1.39) one sees that $(A, B)_\omega := \omega(A^*B)$ defines a pre-inner product on $\mathfrak{A}$. Hence from the Cauchy–Schwarz inequality we obtain the useful bound

$$|\omega(A^*B)|^2 \leq \omega(A^*A)\omega(B^*B). \tag{1.48}$$

Proposition 1.4.2. *A linear map* $\omega : \mathfrak{A} \to \mathbb{C}$ *on a unital* C^**-algebra is positive iff* ω *is bounded and* $\|\omega\| = \omega(\mathbb{I})$. *Hence a state* ω *on a unital* C^**-algebra may* *equivalently be characterized as a positive linear functional for which $\omega(\mathbb{I}) = 1$ or as a bounded linear functional for which $\|\omega\| = \omega(\mathbb{I}) = 1$.*

A state ω on a C^*-algebra without unit has a unique extension to a state $\omega_{\mathbb{I}}$ on the unitization $\mathfrak{A}_{\mathbb{I}}$, given by

$$\omega_{\mathbb{I}}(A + \lambda\mathbb{I}) := \omega(A) + \lambda. \tag{1.49}$$

When ω is positive and $A = A^*$ we have, using (1.40), the bound $|\omega(A)| \leq \omega(\mathbb{I})\|A\|$. For general A the same inequality follows from (1.48) with $A = \mathbb{I}$, (1.15), and the bound just derived. The upper bound is reached by $A = \mathbb{I}$.

To prove the converse claim, we first note that the argument after (1.33) may be copied, showing that ω is real on $\mathfrak{A}_{\mathbb{R}}$. Next, we show that $A \geq 0$ implies $\omega(A) \geq 0$. Choose $s > 0$ small enough so that $\|\mathbb{I} - sA\| \leq 1$. For $\omega \neq 0$ one has $\|\mathbb{I} - sA\| \geq |\omega(\mathbb{I} - sA)|/\omega(\mathbb{I})$, so that $|\omega(\mathbb{I}) - s\omega(A)| \leq \omega(\mathbb{I})$. This is only possible when $\omega(A) \geq 0$.

As to the positivity of $\omega_{\mathbb{I}}$, we observe that $|\omega(A - A\mathbb{I}_{\lambda})| \to 0$ for any approximate unit in $\mathfrak{A}$. Using (1.48) with $B = \mathbb{I}_{\lambda}$, this leads to $|\omega(A)|^2 \leq \omega(A^*A)$. Combining this inequality with (1.47), the definition (1.49) leads to $\omega_{\mathbb{I}}((A + \lambda\mathbb{I})^*(A + \lambda\mathbb{I})) \geq |\omega(A) + \overline{\lambda}|^2 \geq 0$. Hence ω is positive by (1.39). ∎

An important feature of a state space $\mathcal{S}(\mathfrak{A})$ is that it is a convex set. (A convex set $\mathcal{C}$ in a vector space $\mathcal{V}$ is a subset of $\mathcal{V}$ such that the convex sum $\lambda v + (1 - \lambda)w$ belongs to $\mathcal{C}$ whenever $v, w \in \mathcal{C}$ and $\lambda \in [0, 1]$. Geometrically, this means that the line segment between any two points in $\mathcal{C}$ lies in $\mathcal{C}$. It follows that a finite sum $\sum_i p_i v_i$ belongs to $\mathcal{C}$ when all $p_i \geq 0$ and $\sum_i p_i = 1$, and all $v_i \in \mathcal{C}$.) In the unital case it is clear that $\mathcal{S}(\mathfrak{A})$ is convex, since both positivity and normalization are clearly preserved under convex sums. In the nonunital case one arrives at this conclusion most simply via (1.49).

Let $\mathcal{S}(\mathfrak{A})$ be the state space of a unital C^*-algebra $\mathfrak{A}$. Each element ω of $\mathcal{S}(\mathfrak{A})$ is continuous, so that $\mathcal{S}(\mathfrak{A}) \subset \mathfrak{A}^*$. Since w^*-limits obviously preserve positivity and normalization, we see that $\mathcal{S}(\mathfrak{A})$ is closed in $\mathfrak{A}^*$ if the latter is equipped with the w^*-topology. Moreover, $\mathcal{S}(\mathfrak{A})$ is a closed subset of the unit ball of $\mathfrak{A}^*$, so that $\mathcal{S}(\mathfrak{A})$ is compact in the relative w^*-topology by the Banach–Alaoglu theorem. It follows that the state space of a unital C^*-algebra is a compact convex set.

The very simplest example is $\mathfrak{A} = \mathbb{C}$, in which case $\mathcal{S}(\mathfrak{A})$ is a point. The next case is $\mathfrak{A} = \mathbb{C} \oplus \mathbb{C} = \mathbb{C}^2$. The dual is $\mathbb{C}^2$ as well, so that each element of $\left(\mathbb{C}^2\right)^*$ is of the form $\omega(\lambda \dot{+} \mu) = c_1\lambda_1 + c_2\lambda_2$. Positive elements of $\mathbb{C} \oplus \mathbb{C}$ are of the form $\lambda \dot{+} \mu$ with $\lambda \geq 0$ and $\mu \geq 0$, so that a positive functional must have $c_1 \geq 0$ and $c_2 \geq 0$. Since $\mathbb{I} = 1 \dot{+} 1$, normalization yields $c_1 + c_2 = 1$. Identifying 0 with the functional mapping $\lambda \dot{+} \mu$ to λ, and 1 with the one mapping it to μ, we conclude that $\mathcal{S}(\mathbb{C} \oplus \mathbb{C})$ may be identified with the interval $[0, 1]$.

Now consider $\mathfrak{A} = \mathfrak{M}_2(\mathbb{C})$. We identify $\mathfrak{M}_2(\mathbb{C})$ with its dual through the pairing $\omega(A) = \mathrm{Tr}\,\omega A$. It follows that $\mathcal{S}(\mathfrak{A})$ consists of all positive 2×2 matrices ρ with $\mathrm{Tr}\,\rho = 1$; these are the density matrices of quantum mechanics. To identify $\mathcal{S}(\mathfrak{A})$

with a familiar compact convex set, we parametrize

$$\rho = \frac{1}{2}\begin{pmatrix} 1+x & y+iz \\ y-iz & 1-x \end{pmatrix}, \tag{1.50}$$

where $x, y, z \in \mathbb{R}$. The positivity of this matrix then corresponds to the constraint $x^2 + y^2 + z^2 \leq 1$. Hence $\mathcal{S}(\mathfrak{M}_2(\mathbb{C}))$ is the unit ball in $\mathbb{R}^3$.

There are lots of states:

Proposition 1.4.3. *For every $A \in \mathfrak{A}$ and $a \in \sigma(A)$ there is a state ω_a on $\mathfrak{A}$ for which $\omega(A) = a$. When $A = A^*$ there exists a state ω such that $|\omega(A)| = \|A\|$.*

If necessary we add a unit to $\mathfrak{A}$; in the present context this is justified by (1.49). Define a linear map $\tilde{\omega}_a : \mathbb{C}A \oplus \mathbb{C}\mathbb{I} \to \mathbb{C}$ by $\tilde{\omega}_a(\lambda A + \mu\mathbb{I}) := \lambda a + \mu$. Since $a \in \sigma(A)$, one has $\lambda a + \mu \in \sigma(\lambda A + \mu\mathbb{I})$; this easily follows from the definition of $\sigma(A)$. In any Banach algebra one has $r(A) \leq \|A\|$; applying this with A replaced by $\lambda A + \mu\mathbb{I}$ implies $|\tilde{\omega}_a(\lambda A + \mu\mathbb{I})| \leq \|\lambda A + \mu\mathbb{I}\|$. Since $\tilde{\omega}_a(\mathbb{I}) = 1$, it follows that $\|\tilde{\omega}\| = 1$. By the Hahn–Banach theorem there exists an extension ω_a of $\tilde{\omega}$ to $\mathfrak{A}$ of norm 1. Since $\|\omega_a\| = \omega_a(\mathbb{I}) = 1$, this extension is a state, which clearly satisfies $\omega_a(A) = \tilde{\omega}_a(A) = a$.

Since $\sigma(A)$ is closed, there is an $a \in \sigma(A)$ for which $r(A) = |a|$. For this a, and $A = A^*$, one has $|\omega(A)| = |a| = r(A) = \|A\|$ by (1.26); cf. 1.2.4.4. ■

Corollary 1.4.4. *For all $A \in \mathfrak{A}_{\mathbb{R}}$ one has*

$$\|A\| = \sup\{|\omega(A)| \mid \omega \in \mathcal{S}(\mathfrak{A})\}. \tag{1.51}$$

Hence if $\omega(A) = 0$ for all states $\omega \in \mathcal{S}(\mathfrak{A})$, then $A = 0$.

Our goal is to give a geometric realization of a unital C^*-algebra $\mathfrak{A}$ as a certain function space, somewhat in the spirit of Theorem 1.2.3. A function f on a convex set K is called **affine** if it preserves convexity, that is, if

$$f(\lambda\omega_1 + (1-\lambda)\omega_2) = \lambda f(\omega_1) + (1-\lambda) f(\omega_2) \; \forall \omega_1, \omega_2 \in K, \lambda \in [0, 1]. \tag{1.52}$$

The space $A(K, \mathbb{R})$ of all real-valued affine continuous functions on a compact convex set K has a positive cone $A(K, \mathbb{R})^+$, consisting of all positive functions (cf. 1.3.1). Equivalently, $A(K, \mathbb{R})^+$ has a linear partial ordering, in which $f \leq g$ when $f(\omega) \leq g(\omega)$ for all $\omega \in K$. Also, $A(K, \mathbb{R})^+$ is a Banach space in the sup-norm in the case that K is Hausdorff.

Theorem 1.4.5. *The self-adjoint part $\mathfrak{A}_{\mathbb{R}}$ of a unital C^*-algebra $\mathfrak{A}$ is isomorphic as a partially ordered Banach space to the space $A(\mathcal{S}(\mathfrak{A}), \mathbb{R})$ of all real-valued affine continuous functions on the state space $\mathcal{S}(\mathfrak{A})$ of $\mathfrak{A}$ (equipped with the relative w^*-topology).*

The isomorphism in question is given by (1.30), now seen as a map from $\mathfrak{A}$ to the space of functions on the state space $\mathcal{S}(\mathfrak{A})$. It is immediate that this transform is injective. It is a well-known fact in functional analysis that a Banach space $\mathfrak{A}_{\mathbb{R}}$ may be identified under (1.30) with the subspace of its double dual $\mathfrak{A}_{\mathbb{R}}^{**}$ consisting of all w^*-continuous linear functionals on $\mathfrak{A}_{\mathbb{R}}^*$. Since linear functionals on $\mathfrak{A}^*$ are automatically affine on $\mathcal{S}(\mathfrak{A})$, and since we know that a state is real-valued on $\mathfrak{A}_{\mathbb{R}}$, the transform (1.30) maps $\mathfrak{A}_{\mathbb{R}}$ into $A(\mathcal{S}(\mathfrak{A}), \mathbb{R})$. Reinterpreting (1.51) as the equality $\|A\| = \|\hat{A}\|_\infty$, it follows that the map $A \mapsto \hat{A}$ is isometric.

Without proof we state that any Hermitian functional $\varphi \in \mathfrak{A}^*$ on a C^*-algebra $\mathfrak{A}$ has a unique decomposition $\varphi = t_1\omega_1 - t_2\omega_2$, where $\omega_i \in \mathcal{S}(\mathfrak{A})$ and $t_i \in \mathbb{R}^+$. This implies that an element f of $A(\mathcal{S}(\mathfrak{A}), \mathbb{R})$ has a unique extension to a Hermitian linear functional $\tilde{f}$ on $\mathfrak{A}^*$, which is evidently w^*-continuous, and is therefore given by an element $A \in \mathfrak{A}$ (cf. the preceding paragraph). The function $\hat{A}$ in (1.30) is evidently f, so that the image of $\mathfrak{A}$ under $A \mapsto \hat{A}$ is all of $A(\mathcal{S}(\mathfrak{A}), \mathbb{R})$.

Finally, it is trivial from the pertinent definitions that the transform (1.30) preserves positivity. □

As promised, we now complete the proof of Theorem 1.1.9. We discuss the unital case; the nonunital case may be reduced to this, using 1.2.1.

For the first half it remained to be shown that the norm on the self-adjoint part of a C^*-algebra satisfies (1.10). This follows from the order inequality $A^2 \leq A^2 + B^2$ (which is derived from (1.38) and the linearity of the partial ordering) and (1.41).

In the second half we need to prove that (1.25) defines a norm on $\mathfrak{A}$. Firstly, the property $\|A\| = 0 \Rightarrow A = 0$ follows from Lemma 1.3.5. Secondly, the triangle inequality follows by successively using (1.25) with A replaced by $A + B$, (1.51), (1.48), and again (1.25) and (1.51), this time from right to left. Finally, we use (1.42); taking the norm in $\mathfrak{A}_{\mathbb{R}}$ and using (1.25) yields (1.14). ■

Thus from now on a JLB-algebra and the self-adjoint part of a C^*-algebra will be one and the same object.

1.5 Representations and the GNS-Construction

In the theory of C^*-algebras, Hilbert spaces are most naturally regarded as modules, and the material of this section explains how the usual Hilbert space framework of quantum mechanics emerges from the algebraic setting.

Definition 1.5.1. *A* **representation** *of a C^*-algebra $\mathfrak{A}$ on a Hilbert space $\mathcal{H}$ is a morphism $\pi : \mathfrak{A} \to \mathfrak{B}(\mathcal{H})$.*

From the Jordan–Lie point of view, this means that $\pi : \mathfrak{A}_{\mathbb{R}} \to \mathfrak{B}(\mathcal{H})_{\mathbb{R}}$ is a morphism of Jordan–Lie algebras (cf. 1.1.3); here the Jordan–Lie structure on both spaces is given by (1.22). In view of 1.3.10.2 a representation π is automatically continuous; hence $\|\pi(A)\| \leq \|A\|$ for all $A \in \mathfrak{A}$. When π is faithful, this sharpens to $\|\pi(A)\| = \|A\|$ by 1.3.10.3.

There is a natural equivalence relation in the set of all representations of $\mathfrak{A}$: Two representations π_1, π_2 on Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, respectively, are called **equivalent** if there exists a unitary isomorphism $U : \mathcal{H}_1 \to \mathcal{H}_2$ such that $U\pi_1(A)U^* = \pi_2(A)$ for all $A \in \mathfrak{A}$.

The map $\pi(A) = 0$ for all $A \in \mathfrak{A}$ is a representation; more generally, such trivial π may occur as a summand. To exclude this possibility, one says that a representation is **nondegenerate** if 0 is the only vector annihilated by all representatives of $\mathfrak{A}$.

A representation π is called **cyclic** if its carrier space $\mathcal{H}$ contains a **cyclic vector** Ω for π; this means that the closure of $\pi(\mathfrak{A})\Omega$ (which in any case is a closed subspace of $\mathcal{H}$) coincides with $\mathcal{H}$.

Proposition 1.5.2. *Any nondegenerate representation π is a direct sum of cyclic representations.*

The proof uses a lemma that appears in many other proofs as well.

Lemma 1.5.3. *Let $\mathfrak{M}$ be a *-algebra in $\mathfrak{B}(\mathcal{H})$, Ψ a nonzero vector $\mathcal{H}$, and p the projection onto the closure of $\mathfrak{M}\Psi$. Then $p \in \mathfrak{M}'$ (that is, $[p, A] = 0$ for all $A \in \mathfrak{M}$).*

If $A \in \mathfrak{M}$, then $Ap\mathcal{H} \subseteq p\mathcal{H}$ by definition of p. Hence $p^{\perp}Ap = 0$ with $p^{\perp} = \mathbb{I} - p$. When $A = A^*$ this yields $[A, p] = 0$; by (1.21) this is true for all $A \in \mathfrak{M}$. ■

Apply this lemma with $\mathfrak{M} = \pi(\mathfrak{A})$; the assumption of nondegeneracy guarantees that p is nonzero, and the conclusion implies that $A \mapsto p\pi(A)$ defines a cyclic subrepresentation of $\mathfrak{A}$ on $p\mathcal{H}$. This process may be repeated on $p^{\perp}\mathcal{H}$, etc. ■

If π is a nondegenerate representation of a C^*-algebra $\mathfrak{A}$ on $\mathcal{H}$, then any unit vector $\Omega \in \mathcal{H}$ defines a state $\omega \in \mathcal{S}(\mathfrak{A})$, referred to as a **vector state** relative to π, by means of (1.46). Conversely, from any state $\omega \in \mathcal{S}(\mathfrak{A})$ on $\mathfrak{A}$ one can construct a cyclic representation π_ω on a Hilbert space $\mathcal{H}_\omega$ with cyclic vector Ω_ω in the following way. We restrict ourselves to the unital case; the general case follows by adding a unit to $\mathfrak{A}$ and extending ω to $\mathfrak{A}_{\mathbb{I}}$ by (1.49).

Construction 1.5.4.

1. *Given $\omega \in \mathcal{S}(\mathfrak{A})$, define the sesquilinear form $(\,,\,)_0^\omega$ on $\mathfrak{A}$ by*
$$(A, B)_0^\omega := \omega(A^*B). \tag{1.53}$$
Since ω is a state, hence a positive functional, this form is positive semidefinite (this means that $(A, A)_0^\omega \geq 0$ for all A). Its null space
$$\mathcal{N}_\omega = \{A \in \mathfrak{A} \mid \omega(A^*A) = 0\} \tag{1.54}$$
is a left ideal in $\mathfrak{A}$.
2. *The form $(\,,\,)_0^\omega$ projects to an inner product $(\,,\,)_\omega$ on the quotient $\mathfrak{A}/\mathcal{N}_\omega$. If $V_\omega : \mathfrak{A} \to \mathfrak{A}/\mathcal{N}_\omega$ is the canonical projection, then by definition*
$$(V_\omega A, V_\omega B)_\omega := (A, B)_0^\omega. \tag{1.55}$$
The Hilbert space $\mathcal{H}_\omega$ is the closure of $\mathfrak{A}/\mathcal{N}_\omega$ in this inner product.
3. *The representation $\pi_\omega(\mathfrak{A})$ is firstly defined on $\mathfrak{A}/\mathcal{N}_\omega \subset \mathcal{H}_\omega$ by*
$$\pi_\omega(A)V_\omega B := V_\omega AB; \tag{1.56}$$
it follows that π_ω is continuous. Hence $\pi_\omega(A)$ may be defined on all of $\mathcal{H}_\omega$ by continuous extension of (1.56).

4. *The cyclic vector is defined by* $\Omega_\omega = V_\omega \mathbb{I}$, *so that*

$$(\Omega_\omega, \pi_\omega(A)\Omega_\omega) = \omega(A) \quad \forall A \in \mathfrak{A}. \tag{1.57}$$

We now prove the various claims made here. First note that the null space $\mathcal{N}_\omega$ of $(\ ,\ )_0^\omega$ can be defined in two equivalent ways, since

$$\mathcal{N}_\omega := \{A \in \mathfrak{A} \mid (A, A)_0^\omega = 0\} = \{A \in \mathfrak{A} \mid (A, B)_0^\omega = 0 \,\forall\, B \in \mathfrak{A}\}. \tag{1.58}$$

The equivalence follows from (1.48). With the continuity of ω, the equality (1.58) implies that $\mathcal{N}_\omega$ is a left ideal. Hence π_ω in (1.56) is well-defined on the dense subspace $\mathfrak{A}/\mathcal{N}_\omega \subset \mathcal{H}_\omega$, where it clearly satisfies (1.19), with $\varphi \to \pi_\omega$. Also, (1.20) may be verified from (1.55) and (1.53).

To prove that π_ω is bounded on $\mathfrak{A}/\mathcal{N}_\omega$, we compute $\|\pi_\omega(A)\Psi\|^2$ for $\Psi = V_\omega B$, where $A, B \in \mathfrak{A}$. From (1.55) and (1.53) one has $\|\pi_\omega(A)\Psi\|^2 = \omega(B^*A^*AB)$. By (1.42) and the positivity of ω one has $\omega(B^*A^*AB) \leq \|A\|^2\omega(B^*B)$. But $\omega(B^*B) = \|\Psi\|^2$, so that $\|\pi_\omega(A)\| \leq \|A\|$. Hence π_ω may be extended to all of $\mathcal{H}_\omega$, where (1.55) and (1.53) hold by continuity.

Proposition 1.5.5. *If a representation $\pi(\mathfrak{A})$ on $\mathcal{H}$ is cyclic, then the GNS-representation $\pi_\omega(\mathfrak{A})$ on $\mathcal{H}_\omega$ defined by any vector state Ω (corresponding to a cyclic unit vector $\Omega \in \mathcal{H}$) is equivalent to $\pi(\mathfrak{A})$.*

The operator $U : \mathcal{H}_\omega \to \mathcal{H}$ implementing the equivalence is initially defined on the dense subspace $\pi_\omega(\mathfrak{A})\Omega_\omega$ by $U\pi_\omega(A)\Omega_\omega = \pi(A)\Omega$; this operator is well-defined, for $\pi_\omega(A)\Omega_\omega = 0$ implies $\pi(A)\Omega = 0$ by the GNS-construction. It follows from (1.57) that U is unitary as a map from $\mathcal{H}_\omega$ to $U\mathcal{H}_\omega$, but since Ω is cyclic for π, the image of U is $\mathcal{H}$. Hence U is unitary. One verifies that U intertwines π_ω and π. ■

Corollary 1.5.6. *If the Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ of two cyclic representations π_1, π_2 each contain a cyclic vector $\Omega_1 \in \mathcal{H}_1$, $\Omega_2 \in \mathcal{H}_2$ such that*

$$\omega_1(A) := (\Omega_1, \pi_1(A)\Omega_1) = (\Omega_2, \pi_2(A)\Omega_2) =: \omega_2(A)$$

for all $A \in \mathfrak{A}$, then $\pi_1(\mathfrak{A})$ and $\pi_2(\mathfrak{A})$ are equivalent.

By 1.5.5 the representation π_1 is equivalent to the GNS-representation π_{ω_1}, and π_2 is equivalent to π_{ω_2}. On the other hand, π_{ω_1} and π_{ω_2} are induced by the same state $\omega_1 = \omega_2$, so they must coincide. ■

The state ω is called **faithful** when its GNS-representation π_ω is faithful. This is guaranteed when $\mathcal{N}_\omega = 0$, but note that even in that case $\mathcal{H}_\omega$ does not coincide with $\mathfrak{A}$, as the topology on $\mathfrak{A}$ in the operator-norm is finer than the topology of the norm $\|A\|_\omega^2 := (A, A)_0^\omega$.

The GNS-construction leads to a simple proof of Theorem 1.1.8, which uses the following notion.

Definition 1.5.7. *The* **universal representation** *π_u of a C^*-algebra $\mathfrak{A}$ is the direct sum of all its GNS-representations π_ω, $\omega \in \mathcal{S}(\mathfrak{A})$; hence it is defined on the Hilbert space $\mathcal{H}_u = \oplus_{\omega \in \mathcal{S}(\mathfrak{A})}\mathcal{H}_\omega$.*

Theorem 1.1.8 then follows by taking $\mathcal{H} = \mathcal{H}_{\mathrm{u}}$; the desired isomorphism is π_{u}. If $\pi_{\mathrm{u}}(A) = 0$ for some $A \in \mathfrak{A}$, then $\pi_\omega(A)\Omega_\omega = 0$ for all states ω, whence $\|\pi_\omega(A)\Omega_\omega\|^2 = \omega(A^*A) = 0$ by the GNS-construction, so that $A^*A = 0$ by 1.4.3, and finally $A = 0$ by (1.15). Hence π_{u} is faithful, and therefore isometric by 1.3.10.3. ■

1.6 Examples of C^*-Algebras and State Spaces

In this section we give some elementary examples of C^*-algebras.

Example 1.6.1. *Commutative C^*-algebras*

Let X be a discrete space. Take $\mathfrak{A} := \ell_0(X)$, which is the closure (in the sup-norm) of $\ell_c(X)$. The space $\ell_0(X)$ is a C^*-algebra under pointwise multiplication and complex conjugation; see 1.2. By elementary Banach space theory, the dual of $\mathfrak{A}$ is $\mathfrak{A}^* = \ell^1(X)$ under the pairing $\rho(f) = \operatorname{Tr} \rho f := \sum_{x \in X} \rho(x) f(x)$. The positive cone in $\mathfrak{A}$ or $\mathfrak{A}^*$ consists of the positive functions f or ρ. The state space $\mathcal{S}(\mathfrak{A})$ is the set of those positive functions ρ for which $\operatorname{Tr} \rho = \sum_x \rho(x) = 1$.

For example, for a given $y \in X$, the function $\rho = \delta_y$, defined by $\delta_y(x) := \delta_{xy}$, is a state; one clearly has $\delta_y(f) = f(y)$. Hence by Corollary 1.5.6 the one-dimensional representation π_y, defined on $\mathcal{H}_y = \mathbb{C}$ by $\pi_y(f) := f(y)$, is equivalent to the GNS-representation π_{δ_y} (the pertinent cyclic vector in $\mathbb{C}$ is simply $\Omega = 1$).

A positive normalized function on X defines a faithful state when it is strictly positive on X. The GNS-representation $\pi_\rho(\ell_0(X))$ of a faithful state ρ is equivalent to the representation π on $\mathcal{H} = \ell^2(X)$ (with counting measure) by multiplication operators, i.e., $\pi(f)\Psi(x) := f(x)\Psi(x)$. To see this, we first write the inner product in $\ell^2(X)$ as $(\Psi, \Phi) = \operatorname{Tr} \Psi^* \Phi := \sum_x \overline{\Psi(x)}\Phi(x)$. Then note that since $\operatorname{Tr} \rho = 1$, one has $\rho^{1/2} \in \ell^2(X)$. It is clear from the property $\rho(x) > 0$ for all x that $\rho^{1/2}$ is a cyclic vector for $\pi(\ell_0(X))$, with the property $(\rho^{1/2}, \pi(f)\rho^{1/2}) = \rho(f)$ for all $f \in \ell_0(X)$. The equivalence between π_ρ and π then follows from Corollary 1.5.6.

Adding the fact that the double dual of $\mathfrak{A}$ is $\ell_0(X)^{**} = \ell^\infty(X)$, we summarize the situation by

$$\ell_c(X) \subseteq \ell^1(X) = \ell_0(X)^* \subseteq \ell^2(X) \subseteq \ell_0(X) \subseteq \ell^\infty(X) = \ell_0(X)^{**}. \tag{1.59}$$

When X is finite all inclusions are replaced by equalities; when X is infinite all inclusions are strict.

Now take X to be a locally compact Hausdorff space, and put $\mathfrak{A} := C_0(X)$ with the sup-norm; this is the closure of $C_c(X)$. Recall that a **Radon measure** is a Borel measure that is inner regular with respect to compact sets. By the Riesz representation theorem, $\mathfrak{A}^*$ is the space of all complex Radon measures μ on X with finite total mass $\mu(X)$. With $\mathfrak{A}^+$ consisting of the positive functions in $\mathfrak{A}$, the dual cone $\mathfrak{A}^{*+}$ is the subspace of $\mathfrak{A}^*$ of nonnegative finite Radon measures. The state space $\mathcal{S}(\mathfrak{A}) = M_1^+(X)$ then consists of the probability measures on X. The GNS-representation π_μ of a state $\mu \in \mathcal{S}(\mathfrak{A})$ is realized on $\mathcal{H}_\mu = L^2(X, \mu)$, on which $\pi_\mu(f)$ is f as a multiplication operator.

Example 1.6.2. *Noncommutative C^*-algebras*

When a Hilbert space $\mathcal{H} = \mathbb{C}^N$ is finite-dimensional, the "maximally noncommutative" C^*-algebra of operators on $\mathcal{H}$ is the algebra $\mathfrak{M}_N(\mathbb{C})$ of $N \times N$ matrices. The appropriate generalization of $\mathfrak{M}_N(\mathbb{C})$ to the infinite-dimensional case is as follows. A **projection** in a *-algebra is an element satisfying $p^2 = p^* = p$.

Definition 1.6.3. *Let $\mathcal{H}$ be a Hilbert space. The *-algebra $\mathfrak{B}_f(\mathcal{H})$ of* **finite-rank operators on** *$\mathcal{H}$ is the (finite) linear span of all finite-dimensional projections on $\mathcal{H}$. In other words, an operator $A \in \mathfrak{B}(\mathcal{H})$ lies in $\mathfrak{B}_f(\mathcal{H})$ when $A\mathcal{H} := \{A\Psi \mid \Psi \in \mathcal{H}\}$ is finite-dimensional.*

The **C^*-algebra $\mathfrak{B}_0(\mathcal{H})$ of compact operators** *on $\mathcal{H}$ is the norm-closure of $\mathfrak{B}_f(\mathcal{H})$ in $\mathfrak{B}(\mathcal{H})$ (with all C^*-algebraic operations borrowed from $\mathfrak{B}(\mathcal{H})$).*

It is clear that $\mathfrak{B}_f(\mathcal{H})$ is a *-algebra, since $p^* = p$ for any projection p. It is obvious that $\mathfrak{B}_f(\mathcal{H})$ is closed under right multiplication by elements of $\mathfrak{B}(\mathcal{H})$; since it is a *-algebra, it is therefore also closed under left multiplication. By continuity of multiplication in $\mathfrak{B}(\mathcal{H})$, it follows that $\mathfrak{B}_0(\mathcal{H})$ is an ideal in $\mathfrak{B}(\mathcal{H})$. It is easily verified that the unit operator $\mathbb{I}$ lies in $\mathfrak{B}_0(\mathcal{H})$ iff $\mathcal{H}$ is finite-dimensional.

We know from the theory of single operators on a Hilbert space that the image of the unit ball in $\mathcal{H}$ under an element $A \in \mathfrak{B}_0(\mathcal{H})$ is compact (in the strong topology on $\mathcal{H}$); this explains the name of $\mathfrak{B}_0(\mathcal{H})$. A self-adjoint operator $A \in \mathfrak{B}(\mathcal{H})$ is compact iff $A = \sum_i a_i[\Psi_i]$ (norm-convergent sum), where each eigenvalue a_i has finite multiplicity, and $\lim_{i\to\infty} |a_i| = 0$ (where the eigenvalues have been ordered so that $a_i \leq a_j$ when $i > j$). In other words, the set of eigenvalues is discrete, and can have only 0 as a possible accumulation point.

We now wish to determine the state space of $\mathfrak{B}_0(\mathcal{H})$. This involves the study of a number of other subspaces of $\mathfrak{B}(\mathcal{H})$, whose definition we recall.

Definition 1.6.4. *The* **Hilbert–Schmidt norm** *$\|A\|_2$ of $A \in \mathfrak{B}(\mathcal{H})$ is defined by*

$$\|A\|_2^2 := \sum_i \|Ae_i\|^2 = \mathrm{Tr}\,(|A|^2), \tag{1.60}$$

where $\{e_i\}_i$ is an arbitrary basis of $\mathcal{H}$; the right-hand side is independent of the choice of basis. Also, $|A| := \sqrt{A^*A}$ is defined by the continuous functional calculus. The **Hilbert–Schmidt class** *$\mathfrak{B}_2(\mathcal{H})$ consists of all $A \in \mathfrak{B}(\mathcal{H})$ for which $\|A\|_2 < \infty$.*

The **trace norm** *$\|A\|_1$ of $A \in \mathfrak{B}(\mathcal{H})$ is defined by*

$$\|A\|_1 := \||A|^{1/2}\|_2^2 = \mathrm{Tr}\,|A|. \tag{1.61}$$

The **trace class** *$\mathfrak{B}_1(\mathcal{H})$ consists of all $A \in \mathfrak{B}(\mathcal{H})$ for which $\|A\|_1 < \infty$.*

The noncommutative analogue of (1.59) is as follows.

Theorem 1.6.5. *One has the inclusions*

$$\mathfrak{B}_f(\mathcal{H}) \subseteq \mathfrak{B}_1(\mathcal{H}) = \mathfrak{B}_0(\mathcal{H})^* \subseteq \mathfrak{B}_2(\mathcal{H}) \subseteq \mathfrak{B}_0(\mathcal{H}) \subseteq \mathfrak{B}(\mathcal{H}) = \mathfrak{B}_0(\mathcal{H})^{**}, \tag{1.62}$$

where the (isometric) identification of $\mathfrak{B}_1(\mathcal{H})$ *with* $\mathfrak{B}_0(\mathcal{H})^*$ *is made through the pairing*

$$\rho(A) = \operatorname{Tr} \rho A, \tag{1.63}$$

and $\mathfrak{B}(\mathcal{H})$ *is (isometrically) identified with* $\mathfrak{B}_1(\mathcal{H})^* = \mathfrak{B}_0(\mathcal{H})^{**}$ *through the pairing* $A(\rho) = \operatorname{Tr} \rho A$.

The definition of the norms in (1.6.4) easily leads to $\|A\| \leq \|A\|_i$ for $i = 1, 2$. Since $\mathfrak{B}_0(\mathcal{H})$, $\mathfrak{B}_1(\mathcal{H})$, and $\mathfrak{B}_2(\mathcal{H})$ are the completions of $\mathfrak{B}_f(\mathcal{H})$ in the norms $\|\cdot\|$, $\|\cdot\|_1$, and $\|\cdot\|_2$, respectively, these inequalities imply that $\mathfrak{B}_i(\mathcal{H}) \subseteq \mathfrak{B}_0(\mathcal{H})$ for $i = 1, 2$. Using the characterization of self-adjoint compact operators mentioned above, one then infers from 1.6.4 that $\|A\|_2 \leq \|A\|_1$, so that $\mathfrak{B}_1(\mathcal{H}) \subseteq \mathfrak{B}_2(\mathcal{H})$.

The inclusions $\mathfrak{B}_1(\mathcal{H}) \subseteq \mathfrak{B}_0(\mathcal{H})^*$ and $\mathfrak{B}(\mathcal{H}) \subseteq \mathfrak{B}_1(\mathcal{H})^*$ both follow from the (nontrivial) estimate

$$|\operatorname{Tr} \rho A| \leq \|A\| \, \|\rho\|_1. \tag{1.64}$$

To show that $\mathfrak{B}_0(\mathcal{H})^* \subseteq \mathfrak{B}_1(\mathcal{H})$ one restricts a given element $\hat{\rho} \in \mathfrak{B}_0(\mathcal{H})^*$ to $\mathfrak{B}_2(\mathcal{H})$, on which it is continuous. Now, the operator space $\mathfrak{B}_2(\mathcal{H})$ is a Hilbert space in the inner product $(A, B) := \operatorname{Tr} A^* B$, so that by Riesz–Fischer there must be an operator $\rho \in \mathfrak{B}_2(\mathcal{H})$ such that $\hat{\rho}(A) = \operatorname{Tr} \rho A$ for all $A \in \mathfrak{B}_2(\mathcal{H})$. One then shows that $|\operatorname{Tr} p|\rho|| \leq \|\hat{\rho}\|$ for any finite-dimensional projection p, which implies that $\|\rho\|_1 \leq \|\hat{\rho}\|$, so that $\rho \in \mathfrak{B}_1(\mathcal{H})$. With the opposite inequality from (1.64), this proves that $\mathfrak{B}_1(\mathcal{H}) = \mathfrak{B}_0(\mathcal{H})^*$ isometrically.

To establish the inclusion $\mathfrak{B}_1(\mathcal{H})^* \subseteq \mathfrak{B}(\mathcal{H})$, pick $\hat{A} \in \mathfrak{B}_1(\mathcal{H})^*$, and define a quadratic form Q_A on $\mathcal{H}$ by $Q_A(\Psi, \Phi) := \hat{A}(|\Phi\rangle\langle\Psi|)$. Here the operator $|\Phi\rangle\langle\Psi|$ is defined by $|\Phi\rangle\langle\Psi|\Omega := (\Psi, \Omega)\Phi$. This form is easily seen to be bounded by $\|\hat{A}\|$, so that it is implemented by a bounded operator A, in that $Q_A(\Psi, \Phi) = (\Psi, A\Phi)$. By linear extension to $\mathfrak{B}_f(\mathcal{H})$ and subsequently continuous extension to $\mathfrak{B}_1(\mathcal{H})$, this implies that $\hat{A}(\rho) = \operatorname{Tr} \rho A$, with $\|A\| \leq \|\hat{A}\|$. Since (1.64) implies the opposite inequality, this proves the last claim. □

Corollary 1.6.6. *The state space* $\mathcal{S}(\mathfrak{B}_0(\mathcal{H}))$ *of the* C^**-algebra* $\mathfrak{B}_0(\mathcal{H})$ *of all compact operators on some Hilbert space* $\mathcal{H}$ *consists of all density matrices, where a* **density matrix** *is an element* $\rho \in \mathfrak{B}_1(\mathcal{H})$ *that is positive* ($\rho \geq 0$) *and has unit trace* ($\operatorname{Tr} \rho = 1$)*, and the corresponding state is defined in (1.63).*

Since $\rho \in \mathfrak{B}_1(\mathcal{H})$ is compact, one may diagonalize it by $\rho = \sum_i p_i [\Psi_i]$. Using $A = [\Psi_i]$, which is positive, the condition $\rho(A) \geq 0$ yields $p_i \geq 0$. Conversely, when all $p_i \geq 0$, the operator ρ is positive. The normalization condition $\|\rho\|_1 = \sum p_i = 1$ completes the characterization of $\mathcal{S}(\mathfrak{B}_0(\mathcal{H}))$. ■

Proposition 1.6.7.

1. *For each unit vector* $\Psi \in \mathcal{H}$ *the GNS-representation* $\pi_\psi(\mathfrak{B}_0(\mathcal{H}))$ *corresponding to the density matrix* $\rho = [\Psi]$ *is equivalent to the defining representation.*
2. *The GNS-representation* π_ρ *corresponding to a faithful state* ρ *on* $\mathfrak{B}_0(\mathcal{H})$ *is equivalent to the representation* $\hat{\pi}_\rho(\mathfrak{B}_0(\mathcal{H}))$ *on the Hilbert space* $\mathfrak{B}_2(\mathcal{H})$ *of Hilbert–Schmidt operators given by left multiplication, i.e.,* $\hat{\pi}_\rho(A)B := AB$.

The first claim is immediate from the property $\mathrm{Tr}\,[\Psi]A = (\Psi, A\Psi)$ and 1.5.5. For the second, it is obvious from 1.6.4 that for $A \in \mathfrak{B}(\mathcal{H})$ and $B \in \mathfrak{B}_2(\mathcal{H})$ one has $\|AB\|_2 \leq \|A\|\,\|B\|_2$, so that the representation $\hat{\pi}_\rho$ is well-defined. When $\rho \in \mathfrak{B}_1(\mathcal{H})$ and $\rho \geq 0$, then $\rho^{1/2} \in \mathfrak{B}_2(\mathcal{H})$, and it is easily seen that $\rho^{1/2}$ is cyclic for $\hat{\pi}_\rho(\mathfrak{B}_0(\mathcal{H}))$ when $\hat{\rho}$ is faithful. Using the fact that for $A, B \in \mathfrak{B}_2(\mathcal{H})$ one has $\mathrm{Tr}\,AB = \mathrm{Tr}\,BA$, we compute $(\rho^{1/2}, \hat{\pi}_\rho(A)\rho^{1/2}) = \rho(A)$. The equivalence between π_ρ and $\hat{\pi}_\rho$ then follows from 1.5.6. ■

1.7 Von Neumann Algebras

In this section we state some basic facts about von Neumann algebras (which will be used only as ancillary tools). The **commutant** $\mathfrak{M}'$ of some collection $\mathfrak{M}$ of bounded operators on a Hilbert space is the set of all bounded operators that commute with all elements of $\mathfrak{M}$; the **bicommutant** $\mathfrak{M}''$ is the commutant of $\mathfrak{M}'$. One verifies that $\mathfrak{M}''' = \mathfrak{M}'$. The main result is the so-called **double commutant theorem**, which we will first state in the finite-dimensional case.

Proposition 1.7.1. *Let $\mathcal{H} = \mathbb{C}^n$ be a finite-dimensional Hilbert space, and let $\mathfrak{M}$ be a *-algebra (and hence a C^*-algebra) in $\mathfrak{B}(\mathcal{H}) = \mathfrak{M}_n(\mathbb{C})$ containing $\mathbb{I}$. Then $\mathfrak{M}'' = \mathfrak{M}$.*

Choose some $\Psi \in \mathcal{H}$, form the linear subspace $\mathfrak{M}\Psi$ of $\mathcal{H}$, and consider the projection $p = [\mathfrak{M}\Psi]$ onto this subspace. By Lemma 1.5.3 one has $p \in \mathfrak{M}'$. Hence $A \in \mathfrak{M}''$ commutes with p. Since $\mathbb{I} \in \mathfrak{M}$, we therefore have $\Psi = \mathbb{I}\Psi \in \mathfrak{M}\Psi$, so $\Psi = p\Psi$, and $A\Psi = Ap\Psi = pA\Psi \in \mathfrak{M}\Psi$. Hence $A\Psi = A_0\Psi$ for some $A_0 \in \mathfrak{M}$.

Choose $\Psi_1, \ldots, \Psi_n \in \mathcal{H}$, and regard $\Omega := \Psi_1 \dot{+} \cdots \dot{+} \Psi_n$ as an element of $\mathcal{H}^n := \oplus^n \mathcal{H} \simeq \mathcal{H} \otimes \mathbb{C}^n$ (the direct sum of n copies of $\mathcal{H}$), where Ψ_i lies in the ith copy. Identify $\mathfrak{B}(\mathcal{H}^n)$ with the algebra $\mathfrak{M}_n(\mathfrak{B}(\mathcal{H}))$ of $n \times n$ matrices with entries in $\mathfrak{B}(\mathcal{H})$, and embed $\mathfrak{M}$ in $\mathfrak{M}_n(\mathfrak{B}(\mathcal{H}))$ by $A \mapsto \delta(A) := A\mathbb{I}_n^{\otimes}$, where $\mathbb{I}_n^{\otimes}$ is the unit in $\mathfrak{M}_n(\mathfrak{B}(\mathcal{H}))$; this is the diagonal matrix in $\mathfrak{M}_n(\mathfrak{B}(\mathcal{H}))$ in which all diagonal entries are A.

Now use the first part of the proof, with $\mathcal{H}$, $\mathfrak{M}$, A, and Ψ replaced by $\mathcal{H}^n$, $\delta(\mathfrak{M})$, $\mathbb{A} := \delta(A)$, and Ω, respectively. Hence given $\Psi_1, \ldots, \Psi_n$ and $\delta(A) \in \delta(\mathfrak{M})$ there exists $\mathbb{A}_0 \in \delta(\mathfrak{M})''$ such that $\delta(A)\Omega = \mathbb{A}_0\Omega$. For arbitrary $\mathbb{B} \in \mathfrak{M}_n(\mathfrak{B}(\mathcal{H}))$, compute $([\mathbb{B}, \delta(A)])_{ij} = [B_{ij}, A]$. Hence $\delta(\mathfrak{M})' = \mathfrak{M}_n(\mathfrak{M}')$. It is easy to see that $\mathfrak{M}_n(\mathfrak{M}')' = \mathfrak{M}_n(\mathfrak{M}'')$, so that $\delta(\mathfrak{M})'' = \delta(\mathfrak{M}'')$. Therefore, $\mathbb{A}_0 = \delta(A)_0$ for some $A_0 \in \mathfrak{M}$. Hence $A\Psi_i = A_0\Psi_i$ for all $i = 1, \ldots, n$. Since the Ψ_i were arbitrary, this proves that $A = A_0 \in \mathfrak{M}''$. ■

As it stands, Proposition 1.7.1 is not valid when $\mathfrak{M}_n(\mathbb{C})$ is replaced by $\mathfrak{B}(\mathcal{H})$, where $\dim(\mathcal{H}) = \infty$. To describe the appropriate refinement, we define two locally convex topologies on $\mathfrak{B}(\mathcal{H})$ that are weaker than the norm topology we have been using so far.

The seminorms $p_\Psi(A) := \|A\Psi\|$ define the **strong topology** on $\mathfrak{B}(\mathcal{H})$, so that $A_\lambda \to A$ strongly when $\|(A_\lambda - A)\Psi\| \to 0$ for all $\Psi \in \mathcal{H}$. In the proof of 1.7.2

we will use the fact that a neighborhood basis of A is given by all sets of the form $\{B \in \mathfrak{B}(\mathcal{H}) \mid \|(A - B)\Psi_i\| < \epsilon \text{ for all } i = 1, \ldots, n\}$, where $\epsilon > 0$, $n \in \mathbb{N}$, and $\Psi_1, \ldots, \Psi_n \in \mathcal{H}$.

The **weak topology** on $\mathfrak{B}(\mathcal{H})$ is defined by the seminorms $p_{\Psi,\Phi}(A) := |(\Psi, A\Phi)|$, so that $A_\lambda \to A$ weakly when $|(\Psi, (A_\lambda - A)\Psi)| \to 0$ for all $\Psi \in \mathcal{H}$. The norm topology is stronger than the strong topology, which in turn is stronger than the weak topology.

Theorem 1.7.2. *Let $\mathfrak{M}$ be a *-algebra in $\mathfrak{B}(\mathcal{H})$ containing $\mathbb{I}$. The following are equivalent:*

1. $\mathfrak{M}'' = \mathfrak{M}$.
2. *$\mathfrak{M}$ is closed in the weak operator topology.*
3. *$\mathfrak{M}$ is closed in the strong operator topology.*

It is easily verified from the definition of weak convergence that the commutant $\mathfrak{N}'$ of a *-algebra $\mathfrak{N}$ is always weakly closed. If $\mathfrak{M}'' = \mathfrak{M}$, then $\mathfrak{M} = \mathfrak{N}'$ for $\mathfrak{N} = \mathfrak{M}'$, so that $\mathfrak{M}$ is weakly closed. Hence $1 \Rightarrow 2$. Since the weak topology is weaker than the strong topology, $2 \Rightarrow 3$ is trivial.

To prove $3 \Rightarrow 1$, we adapt the proof of 1.7.1 to the infinite-dimensional situation. Instead of $\mathfrak{M}\Psi$, which may not be closed, we consider its closure $\overline{\mathfrak{M}\Psi}$, so that $p = [\overline{\mathfrak{M}\Psi}]$. Hence $A \in \mathfrak{M}''$ implies $A \in \overline{\mathfrak{M}\Psi}$; in other words, for every $\epsilon > 0$ there is an $A_\epsilon \in \mathfrak{M}$ such that $\|(A - A_\epsilon)\Psi\| < \epsilon$. For $\mathcal{H}^n$ this means that $\|\delta(A - A_\epsilon)\Omega\|^2 < \epsilon^2$. The left-hand side of this inequality equals the sum $\sum_{i=1}^n \|(A - A_\epsilon)\Psi_i\|^2$, so that $\|(A - A_\epsilon)\Psi_i\| < \epsilon$ for all $i = 1, \ldots, n$. It follows that $A_\epsilon \to A$ strongly for $\epsilon \to 0$. Since all $A_\epsilon \in \mathfrak{M}$ and $\mathfrak{M}$ is strongly closed, this implies that $A \in \mathfrak{M}$, so that $\mathfrak{M}'' \subseteq \mathfrak{M}$. Since trivially $\mathfrak{M} \subseteq \mathfrak{M}''$, this proves $3 \Rightarrow 1$. ∎

This theorem is remarkable, for it relates a topological condition ($\mathfrak{M}$ being closed in certain topologies) to an algebraic one ($\mathfrak{M}$ being its own bicommutant). A similar but simpler example of such a theorem states that a linear subspace $\mathcal{K}$ of a Hilbert space is closed iff $\mathcal{K} = \mathcal{K}^{\perp\perp}$ (where $\mathcal{K}^\perp$ is the orthogonal complement of $\mathcal{K}$).

Definition 1.7.3. *A *-algebra $\mathfrak{M}$ (containing the unit operator) of bounded operators on some Hilbert space is called a* **von Neumann algebra** *if it satisfies one (hence all) of the conditions in 1.7.2.*

We know from 1.6.5 that $\mathfrak{B}(\mathcal{H}) = \mathfrak{B}_1(\mathcal{H})^*$; the pertinent w^*-topology on $\mathfrak{B}(\mathcal{H})$ is often called the **σ-weak topology**. This topology is generated by the seminorms $p_\rho(A) := |\mathrm{Tr}\,\rho A|$, and is clearly stronger than the weak topology (but weaker than the norm topology). Hence a von Neumann algebra $\mathfrak{M} \subseteq \mathfrak{B}(\mathcal{H})$ is closed in the (relative) σ-weak topology.

Moreover, a von Neumann algebra $\mathfrak{M}$ is closed in the norm topology (defined by the norm (1.18)) as well, so that it is a C^*-algebra. A state on $\mathfrak{M} \subseteq \mathfrak{B}(\mathcal{H})$ of the form (1.63) for a density matrix ρ (cf. 1.6.6) is called **normal**. The linear span of all normal states in $\mathfrak{M}^*$ is called the **predual** $\mathfrak{M}_*$ of $\mathfrak{M}$. For example, the predual of $\mathfrak{B}(\mathcal{H})$ is $\mathfrak{B}_1(\mathcal{H})$, and more generally one has $\mathfrak{M} = \mathfrak{M}_*^*$ as a Banach space. The set $\mathcal{N}(\mathfrak{M}) := \mathcal{S}(\mathfrak{M}) \cap \mathfrak{M}_*$ of all normal states on $\mathfrak{M}$ is called the **normal state space** of $\mathfrak{M}$.

All von Neumann algebras in this book are of the form $\mathfrak{M} = \pi(\mathfrak{A})''$, where π is a representation of some C^*-algebra $\mathfrak{A}$. In particular, one may take $\pi = \pi_{\mathrm{u}}$; cf. 1.5.7.

Proposition 1.7.4. *The bidual $\mathfrak{A}^{**}$ of a C^*-algebra $\mathfrak{A}$ is isomorphic (as a Banach space) to $\pi_{\mathrm{u}}(\mathfrak{A})''$. Through this isomorphism, $\mathfrak{A}^{**}$ acquires the structure of a von Neumann algebra (and therefore of a C^*-algebra).*

The proof is a highly nontrivial generalization of the proof of 1.6.5. The equality $\mathfrak{B}_0(\mathcal{H})^* = \mathfrak{B}_1(\mathcal{H})$ is now replaced by the fact that $\mathfrak{A}^*$ is the linear span of all functionals of the form $A \mapsto (\Psi, \pi_{\mathrm{u}}(\mathfrak{A})\Phi)$, where $\Psi, \Phi \in \mathcal{H}_{\mathrm{u}}$. This characterization is then used to show that $\mathfrak{A}^*$ is the predual of $\pi_{\mathrm{u}}(\mathfrak{A})''$, so that $\mathfrak{A}^{**} = \pi_{\mathrm{u}}(\mathfrak{A})''$. □

In the context of Theorem 1.4.5, we note that when K is a Hausdorff compact convex set, the bidual of $A(K, \mathbb{R})$ (with sup-norm) is the space $A_b(K, \mathbb{R})$ of all bounded real-valued affine functions on K. Hence for a C^*-algebra $\mathfrak{A}$ one has $\mathfrak{A}^{**}_{\mathbb{R}} \simeq \pi_{\mathrm{u}}(\mathfrak{A})''_{\mathbb{R}} \simeq A_b(\mathcal{S}(\mathfrak{A}), \mathbb{R})$. The predual of $\mathfrak{A}^{**}$ is obviously $\mathfrak{A}^{**}_* = \mathfrak{A}^*$, and the normal state space is $\mathcal{N}(\mathfrak{A}^{**}) = \mathcal{S}(\mathfrak{A})$. More generally, for any von Neumann algebra $\mathfrak{M}$ one has $\mathfrak{M}_{\mathbb{R}} \simeq A_b(\mathcal{N}(\mathfrak{M}), \mathbb{R})$ as partially ordered Banach spaces. This isomorphism maps the σ-weak topology on $\mathfrak{M}_{\mathbb{R}}$ to the topology of pointwise convergence on $A_b(\mathcal{N}(\mathfrak{M}), \mathbb{R})$.

The **center** of a von Neumann algebra $\mathfrak{M}$ is $\mathfrak{M} \cap \mathfrak{M}'$; this is the set of all elements of $\mathfrak{M}$ that commute with every element in the algebra. The following proposition allows one to regard $\pi(\mathfrak{A})''$ as a von Neumann subalgebra of $\mathfrak{A}^{**}$.

Proposition 1.7.5. *If π is a cyclic representation of a C^*-algebra $\mathfrak{A}$, there exists a projection p in the center of $\pi_{\mathrm{u}}(\mathfrak{A})''$ such that $\pi(\mathfrak{A})''$ is isomorphic (as a von Neumann algebra) to $p\pi_{\mathrm{u}}(\mathfrak{A})''$.*

The idea of the proof is that the morphism $\pi \circ \pi_{\mathrm{u}}^{-1}$ from $\pi_{\mathrm{u}}(\mathfrak{A})$ to $\pi(\mathfrak{A})$ is σ-weakly continuous, so that it can be extended to a morphism from $\pi_{\mathrm{u}}(\mathfrak{A})''$ to $\pi(\mathfrak{A})''$. The kernel of this extension is a σ-weakly closed ideal in $\pi_{\mathrm{u}}(\mathfrak{A})''$. It can be shown that a σ-weakly closed ideal in a von Neumann algebra $\mathfrak{M}$ is of the form $q\mathfrak{M}$, where q is a projection in the center of $\mathfrak{M}$. Applying this to the case at hand yields 1.7.5, with $p = \mathbb{I} - q$. □

2 The Structure of Pure State Spaces

2.1 *Pure States and Compact Convex Sets*

In this section we look at a subspace of the state space on a C^*-algebra, which may be interpreted as a quantum analogue of the phase space of a classical system.

Let us return to 1.4. One observes that the compact convex sets one naturally has in mind have a boundary; this particularly applies to the state spaces of the

C^*-algebras $\mathbb{C}$, $\mathbb{C} \oplus \mathbb{C}$, and $\mathfrak{M}_2(\mathbb{C})$. The intrinsic definition of this boundary is as follows.

Definition 2.1.1. *An* **extreme point** *in a convex set K is a member ω of K that can be decomposed as $\omega = \lambda\omega_1 + (1-\lambda)\omega_2$, where $\lambda \in (0,1)$, iff $\omega_1 = \omega_2 = \omega$. The collection $\partial_e K$ of extreme points in K is called the* **extreme boundary** *of K.*

An extreme point in the state space $K = \mathcal{S}(\mathfrak{A})$ of a C^*-algebra $\mathfrak{A}$ is called a **pure state**. *A state that is not pure is called* **mixed**. *We write $\mathcal{P}(\mathfrak{A})$, or simply $\mathcal{P}$, for $\partial_e\mathcal{S}(\mathfrak{A})$, referred to as the* **pure state space** *of $\mathfrak{A}$.*

Thus the single state on $\mathbb{C}$ is pure, the pure states on $\mathbb{C} \oplus \mathbb{C}$ are the points 0 and 1 in $[0, 1]$, and the pure states on $\mathfrak{M}_2(\mathbb{C})$ are the matrices ρ in (1.50) for which $x^2 + y^2 + z^2 = 1$. These are the projections onto one-dimensional subspaces of $\mathbb{C}^2$, and we see that $\mathcal{P}(\mathfrak{M}_2(\mathbb{C}))$ may be identified with the unit sphere in $\mathbb{R}^3$. More generally, one has

Proposition 2.1.2. *The pure state space of $\mathfrak{B}_0(\mathcal{H})$ consists of all one-dimensional projections, so that any pure state on $\mathfrak{B}_0(\mathcal{H})$ is a vector state (1.46) in $\mathcal{H}$.*

This is immediate from 1.6.6, the spectral theorem applied to a density matrix, and 2.1.1. ■

A useful reformulation of the notion of a pure state is as follows.

Proposition 2.1.3. *A state is pure iff $0 \leq \rho \leq \omega$ for a positive functional ρ implies $\rho = t\omega$ for some $t \in \mathbb{R}^+$.*

We assume that $\mathfrak{A}$ is unital; if not, use 1.2.1 and (1.49). For $\rho = 0$ or $\rho = \omega$ the claim is obvious. When ω is pure and $0 \leq \rho \leq \omega$, with $0 \neq \rho \neq \omega$, then $0 < \rho(\mathbb{I}) < 1$, since $\omega - \rho$ is positive; hence $\|\omega - \rho\| = \omega(\mathbb{I}) - \rho(\mathbb{I}) = 1 - \rho(\mathbb{I})$. Hence $\rho(\mathbb{I})$ would imply $\omega = \rho$, whereas $\rho(\mathbb{I}) = 0$ implies $\rho = 0$, contrary to assumption. Hence $\omega_1 := (\omega - \rho)/(1 - \rho(\mathbb{I}))$ and $\omega_2 := \rho/\rho(\mathbb{I})$ are states, and $\omega = \lambda\omega_1 + (1-\lambda)\omega_2$ with $\lambda = 1 - \rho(\mathbb{I})$. Since ω is pure, by 2.1.1 we have $\rho = \rho(\mathbb{I})\omega$.

Conversely, if ω is decomposed as in 2.1.1, then $0 \leq \lambda\omega_1 \leq \omega$, so that $\lambda\omega_1 = t\omega$ by assumption; normalization gives $t = \lambda$, hence $\omega_1 = \omega = \omega_2$, and ω is pure. ■

Here is another example of a pure state space.

Proposition 2.1.4. *The pure state space of the commutative C^*-algebra $C_0(X)$ (equipped with the relative w^*-topology) is homeomorphic to X.*

The case that X is not compact may be reduced to the compact case by passing from $\mathfrak{A} = C_0(X)$ to $\mathfrak{A}_{\mathbb{I}} = C(\tilde{X})$ (where $\tilde{X}$ is the one-point compactification of X); cf. (2.2) below. In view of the proof of Theorem 1.2.3, we then merely need to prove that any pure state on $C(X)$ is multiplicative, and vice versa; $\mathcal{P}(C(X))$ and $\Delta(C(X))$ are both equipped with the relative w^*-topology.

Let $\omega_x \in \Delta(C(X))$ (cf. the proof of 1.2.3), and suppose a functional ρ satisfies $0 \leq \rho \leq \omega_x$. Then $\ker(\omega_x) \subseteq \ker(\rho)$, and $\ker(\omega_x)$ is a maximal ideal, so that

$\ker(\omega_x) = \ker(\rho)$. Since two functionals on any vector space are proportional when they have the same kernel, it follows from 2.1.3 that ω_x is pure.

Conversely, let ω be a pure state, and pick a $g \in C(X)$ with $0 \leq g \leq 1_X$. Define a functional ω_g on $C(X)$ by $\omega_g(f) := \omega(fg)$. Since $\omega(f) - \omega_g(f) = \omega(f(1-g))$, and $0 \leq 1 - g \leq 1_X$, one has $0 \leq \omega_g \leq \omega$. Hence $\omega_g = t\omega$ for some $t \in \mathbb{R}^+$ by 2.1.3. Putting $f = 1_X$ yields $t = \omega(g)$. Since any function is a linear combination of functions g for which $0 \leq g \leq 1_X$, it follows that ω is multiplicative. ■

It could be that a given convex set contains no extreme points at all; think of an open convex cone. When K is compact, this possibility is excluded by a basic theorem in functional analysis, which we state without proof. The **convex hull** $\mathrm{co}(V)$ of a subset V of a vector space is defined by

$$\mathrm{co}(V) := \{\lambda v + (1-\lambda)w \mid v, w \in V, \lambda \in [0,1]\}. \tag{2.1}$$

Theorem 2.1.5. *A compact convex set K embedded in a locally convex vector space is the closure of the convex hull of its extreme points. In other words, $K = \overline{\mathrm{co}}(\partial_e K)$.*

Although the state space of a C^*-algebra $\mathfrak{A}$ without unit (such as $\mathfrak{B}_0(\mathcal{H})$ or $C_0(X)$) is not compact, Theorem 2.1.5 may nonetheless be used. For the pure state space of $\mathfrak{A}$ may be described in terms of the pure state space $\mathcal{P}(\mathfrak{A}_{\mathbb{I}})$ of its unitization $\mathfrak{A}_{\mathbb{I}}$; cf. 1.2.1. Define a functional ω_∞ by $\omega_\infty(A + \lambda\mathbb{I}) = \lambda$ for all A; this is easily seen to be a pure state on $\mathfrak{A}_{\mathbb{I}}$. Taking (1.49) into account, one obtains a homeomorphism

$$\mathcal{P}(\mathfrak{A}) \simeq \mathcal{P}(\mathfrak{A}_{\mathbb{I}})\backslash\{\omega_\infty\}. \tag{2.2}$$

The extreme boundary $\partial_e K$ of a compact convex set is not necessarily closed, so that the pure state space $\mathcal{P}(\mathfrak{A})$ of a unital C^*-algebra $\mathfrak{A}$, while always a Hausdorff space, is not generally compact. Nonetheless, it is interesting to realize $\mathfrak{A}_{\mathbb{R}}$ as a subspace of $C(\mathfrak{A}_{\mathbb{R}}(\mathcal{P}), \mathbb{R})$, somewhat in the spirit of 1.4.5. To do so, we replace $\Delta(\mathfrak{A})$ in the definition (1.30) of the Gelfand transform by the pure state space of an arbitrary C^*-algebra; cf. 2.1.4.

Definition 2.1.6. *Let $\mathfrak{A}_{\mathbb{R}}$ be the self-adjoint part of a C^*-algebra $\mathfrak{A}$. The* **Gelfand transform** *of $A \in \mathfrak{A}_{\mathbb{R}}$ is the function $\hat{A} : \mathcal{P}(\mathfrak{A}) \to \mathbb{R}$ defined by (1.30). The subspace $\{\hat{A} \mid A \in \mathfrak{A}_{\mathbb{R}}\}$ of $\ell^\infty(\mathcal{P}(\mathfrak{A}), \mathbb{R})$ is denoted by $\hat{\mathfrak{A}}_{\mathbb{R}}$.*

The extension of the Gelfand transform from $\mathfrak{A}_{\mathbb{R}}$ to $\mathfrak{A}$ is useful only for commutative C^*-algebras; in the noncommutative case the first claim below would not hold if $\mathfrak{A}_{\mathbb{R}}$ were replaced by $\mathfrak{A}$.

Theorem 2.1.7. *The Gelfand transform is an isomorphism between $\mathfrak{A}_{\mathbb{R}}$ and $\hat{\mathfrak{A}}_{\mathbb{R}} \subseteq C(\mathcal{P}(\mathfrak{A}), \mathbb{R})$, seen as partially ordered Banach spaces (here the order in $\mathfrak{A}_{\mathbb{R}}$ is defined by 1.3.3 and 1.3.1, whereas the order in $C(\mathcal{P}(\mathfrak{A}), \mathbb{R})$ is defined by the cone of pointwise positive functions).*

The equality $\hat{\mathfrak{A}}_{\mathbb{R}} = C(\mathcal{P}(\mathfrak{A}), \mathbb{R})$ occurs iff $\mathfrak{A}$ is commutative and unital, in which case $\mathcal{P}(\mathfrak{A})$ is closed.